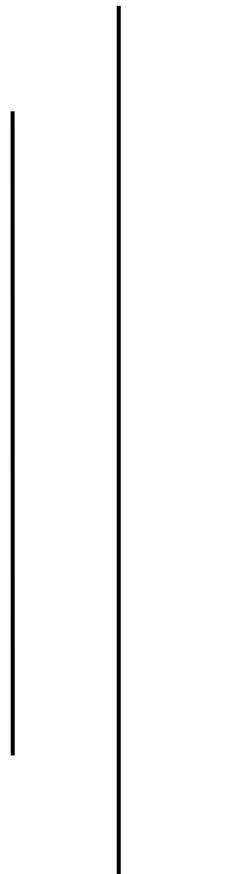


Environmental Management Framework
For
School Sector Reform Program (SSRP)
Nepal



Feb 2015

Acronyms

BPEP	Basic & Primary Education Project
CM	Compliance Monitoring
DEO	District Education Office
DOE	Department of Education
DPs	Development Partners
EA	Environmental Assessment
EfA	Education for All
EIA	Environmental Impact Assessment
EMF	Environmental Management Framework
EMP	Environmental Management Plan
EPR	Environmental Protection Rules
GoN	Government of Nepal
IEE	Initial Environmental Examination
NEGFSIFMN	National Environmental Guidelines for School Improvement and Facility Management in Nepal
NEPAP	National Environmental Policy & Action Plan
PIM	Program Implementation Manual
PSS	Physical Services Section
SESP	Secondary Education Support Program
SMC	School Management Committee
SSRP	School Sector Reform Plan

Executive Summary

School Sector Reform Plan and Construction Works

Physical infrastructure development of schools and District Education Offices (DEO) constitutes an important activity in School Sector Reform Plan (SSRP). Potential environmental issues in SSRP are mainly limited to impacts from construction related works and sites. These, for example, are construction of administrative buildings, schools & hostels, buildings, library & laboratory blocks, construction of new class rooms & rehabilitation of existing class rooms, provisions of toilets & water supply.

Environmental Assessment (EA)

The DOE, in the process of preparing this Environmental Management Framework (EMF), has undertaken an EA for SSRP (Chapter - II). The EA was based on a series of reviews of construction works undertaken in the Basic & Primary Education Project (BPEP II), Secondary Education Support Program (SESP), Education for All (EFA) including field visits and consultation with stakeholders with respect to practices related to school physical facility improvement, and operation and maintenance of facilities. The EA concluded that SSRP's Environmental concerns are related to actions/activities under physical infrastructure.

Environmental issues in SSRP are mainly limited to potential impacts from building construction works and sites. These issues relate to school site selection and planning (appropriateness of sites, orientations of buildings considering climatic factors, and use of appropriate design standards, ensuring supply of safe drinking water (e.g. arsenic-free groundwater in Terai), proper sanitation (school latrine and waste management at school), and appropriate classroom conditions (e.g. light, ventilation, temperature, noise). Location of the school or its facilities in risky or sensitive spots such as in flood-prone/risk, or landslide risk areas, is also a concern besides risks from earthquakes as Nepal is in a high earthquake risk zone.

Minor and localized impacts may arise from construction/upgrading of small scale administrative buildings or school blocks or new/additional classrooms or may be related to providing facilities such as drinking water and sanitation as well as to their operation and maintenance. Since the individual works under SSRP are small, environmental issues are assessed to be small and localized. No large, significant and/or irreversible impact is anticipated. Even cost sharing types of schools, DEO financed schools and laboratory buildings fall into the category of small to medium type construction which doesn't require specific environmental assessment from GON and DPs perspectives. Substantial experiences has been gained from BPEP II, SESP, and EFA to manage these issues.

The EA, therefore, concluded that the SSRP is not likely to present severe environmental risks and/ or highly significant potential adverse impacts in its area of influence. Nevertheless, as the project activities include civil works the EA policy (OP4.01) and IP Policy (4.10) of the World Bank has been triggered to ensure sustainability of resources and no negative impacts on environment during construction and rehabilitation of education units made out of financial support from the World Bank. The project has been categorized as category "B" project. Simultaneously, in order to manage the minor and localized impacts GON has prepared an Environmental Management Framework for SSRP: the EMF has been updated based on the implementation experiences.

Environmental Management Framework (EMF)

The findings of the EA were used to prepare Environmental Management Framework (EMF) for SSRP. This has incorporated the experiences of the other projects such as EFA, emerging lessons and updating the *National Environmental Guidelines for School Improvement and Facility Management in Nepal*.

The EMF is intended to be used as a practical tool during program formulation, design, implementation, and monitoring. The EMF defines simplified steps, procedures and guidelines or criteria and/or standards to be used while planning and developing schools' physical infrastructure under SSRP. These, for example, are related to screening, preparation of simple environmental management and mitigation plan. It describes the steps involved in identifying and mitigating the potential adverse environmental impacts of activities. Hence, EMF in SSRP will ensure the environmentally friendly design and construction of school physical facilities and other educational infrastructures by meeting safeguard policies of the GON. Salient features of the EMF are:

- SSRP activities, in general, do not require formal environmental assessment such as Environmental Impact Assessment (EIA) or Initial Environmental Examination (IEE). However, each SSRP physical infrastructure activity (school improvement activities as well as DEO administrative buildings) will be subject to environmental screening and if required EMP will be prepared.
- At the school level, each participating school will utilize the EMF while preparing the School Improvement Plan (SIP) to consider all the environmental and social issues while planning, designing and construction of education buildings .
- Emphasize effective use of already accepted policy and standards such as National Environmental Guidelines for School Improvement and Facility Management in Nepal; structural and disaster safety of buildings, Public Work Directives, 2002, child- and gender-friendly design and construction, inclusive design and constructions for disables, design guidelines for school buildings construction from environmental perspective (Annex - IX) etc.
 - Recommendations of environmental screening and EMP will be incorporated in the plan, design and contract documents. Monitoring of the construction activities will be done by technical personals (environmental specialist, civil engineers, overseers) from DEO/ DOE or local NGOs. In addition, annual monitoring of ESMF performance will also be done by hiring Independent consultant.
 - The regular monitoring system of the DOE will also include indicators and monitoring mechanisms as specified in the EMF. There will be monitoring during pre-construction, construction and operational phases. DEO will take lead on the capacity building of the school management committees (SMCs) through periodic interactions and orientations to the committees, service providers/design consultant as well as supervision consultant for the monitoring and supervision of construction activities including the environmental and social safeguard compliance.
 -

Institutional Arrangements, Roles and Responsibilities for EMF implementation

The Department of Education (DOE), District Education Office (DEO) and School Management Committee (SMC) are the main implementing agencies. The roles and responsibilities of these

agencies for implementing EMF with respect to planning, construction, and operation of associated facilities have been assigned in the EMF. The School Improvement Plan (SIP) will incorporate the EMF requirements at school level. The school construction will be managed by communities. Some of EMF requirements would have already been incorporated in the School Building Design and Construction Manual and other manuals developed by DOE. The following paragraph presents highlights of the roles and responsibilities of difference agencies:

- DEO will conduct a preliminary survey of environmental conditions of the school and its surrounding area as part of the physical survey of schools.
- The proposed school improvement activities as well as administrative buildings will be subject to environmental screening which will be followed by preparation of EMP only if required. This will be done by the DEO.
- The DOE/DEO ensures incorporation of EMF's planning and design guidelines and provisions (Annex – 9) into SIP. EMP will be prepared for activates such as the construction of district education offices, if required, following the screening process.. EMP will be prepared by the DOE and included as a part of the contract document.
- The Environmental Officer at DOE/PSS will check/ verify random screening and samples of the SIP, and all screening of district education offices. Incorporation of EMP in design/tender documents will be checked by the Environmental Officer of the DOE.
- The DEO regularly monitors/supervises implementation of the environmental works in the schools and reports to DOE. DOE/ PSS Environmental Officer will prepare consolidated report on environmental works. DOE/PSS and the relevant district DEO visits randomly selected samples of schools on four-monthly basis to check environmental compliance as outlined in the EMF. Findings of this will feed into government's quarterly portfolio review meeting.
- .
- DoE will provide overall environmental oversight in implementing EMF, organize awareness/ orientations, manage environmental monitoring etc.. Annual GON and DPs joint review of SSRP will also cover implementation of the EMF. The field level monitoring will be carried out by DEO. Sub project site specific level monitoring will be conducted by DEO at regular basis and by DOE at an intermittent basis. Independent evaluation of EMF performance will be done hiring independent consultant

Capacity Building

Capacity building at different levels (DOE, DEO, and SMC) is necessary in order to implement the EMF successfully. The suggested capacity strengthening measures, for example, include: i) providing environmental competency/human-resources, ii) training, orientation and awareness activities on environmental planning and management of school and school-facilities, and iii) mechanisms for coordination and for accessing specific environmental services e.g. water-quality testing, climate responsive school building construction etc.

In consideration of increasing workload envisaged for implementation of the EMF and the non-existence of environmental competency at centre, GoN/MOE will assign an engineer/technical person / environmental officer at DOE to look after the EMF activities for SSRP. The Officer will be responsible for implementation of the EMF and its provisions, including compliance checking, facilitation, coordination and ensuring dissemination, orientations and capacity buildings activities. Additional human resources or agency will be hired, if necessary, in order to effectively implement the EMF. In addition, DEO will develop and implement an EMF training, orientation and awareness program targeting to SMC, DEO, other district level partners as well as service providers (including engineering design team).

EMF Dissemination and Disclosure

GON will disclose the EMF on their web sites (MOE as well as DOE), and will also place the reports in public places for wider dissemination. Upon approval of EMF by MOE, the DOE will

publish the document in Nepali language. The Nepali version of EMF will be circulated to DEOs and SMCs through the Program Implementation Manual (PIM). The document will be widely circulated to DOE, DEO, collaborating INGOs/NGOs. Through regular orientation program, DOE will disseminate EMF to engineers/sub engineers, district education officers, design consultant and other partners.

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Chapter I - General

1.1 Environmental Management Framework (EMF)

The Government of Nepal has set a goal of creating environmental frameworks for enabling environmental conditions, healthier learning environment in the Nepalese school campuses, but without affecting public health, surrounding environment and communities adversely in the construction and operation of the school physical facilities and other educational infrastructures.

1.2 School Sector Reform Plan (SSRP)

The concept of an integrated grades 1 to 12 school system has been under consideration in Nepal from the 1980s till the Three Year Interim Plan (2007-10). The process of developing and implementing the integrated school system was initiated as early as 1989 with the enactment of the Higher Secondary Education Act. The need for structural adjustment to the integrated school system was further emphasized by Basic and Primary Education Master Plan (1997).

The achievements and experiences gain from the implementation of the EFA sub sector approach and the Secondary Education Support Program (SESP) have encouraged the GON to demonstrate a commitment to embark on School Sector Reform (SSR) with grades 1-12 as an integrated school system in the phased manner from 2009/2010 onwards.

In February 2007, a Concept Paper on SSR was endorsed by the Education Policy Committee. The Concept Paper provided general directions for policy changes and served as the basis for preparation of SSR core documents. As a preparation for an integrated SSR, the core document has clearly spelled out the policy and strategy options for SSR implementation. The preparation activities of SSR will include development of an accelerated implementation/transition plan, illustrating the activities to generate knowledge and to develop local level planning capacity for systemic readiness and resilience.

1.3 EMF in SSRP

The Environmental Management Framework (EMF) is developed to ensure that environmental issues are dealt by in a proper and efficient manner meeting all the legal compliance requirements of GON, and development partners supported SSRP in Nepal.

The EMF will provide an environmental screening process, outline the typical mitigation measures and estimate corresponding budget and capacity building measures for the implementation of SSRP. The EMF is intended to be used as a practical tool during program formulation, design, implementation, and monitoring. It describes the steps involved in identifying and mitigating the potential adverse environmental impacts of activities. EMF ensures the protection of human health, environmental sustainability, and welfare of affected community.

Hence the EMF in SSRP will ensure environmentally friendly design and construction of school physical facilities and other educational infrastructures. In addition, EMF will also safeguard the environmental policy of the GON.

1.4 The Need for EMF in SSRP

EFA through DOE has set the module for the development and implementation of National Environmental Guidelines (NEG) to ensure that the school improvement and facility management under the program meet acceptable national and international norms. This guideline deals with the environmental issues associated with the planning, construction, and operational stages of the school physical facilities and other educational infrastructures, including monitoring the implementation.

Similarly, it is very pertinent to develop, building on the lessons and experiences of the NEG implementation, a comprehensive new version of environmental guideline in the form of an *Environmental Management Framework*, which fits into the objectives of SSR. The lessons learned from the past efforts in implementation of environmental management sector in schools and on the ground implementation of activities through EFA shall be assessed and used in an improved and scientific fashion in SSR while meeting its objectives.

This EMF has been developed by the Ministry of Education of the Government of Nepal especially for the proposed operation to ensure due diligence, to avoid any environmental degradation issues. The purpose of this framework also includes assisting the implementing agency in screening all the sub projects for their likely environmental impacts, identifying suitable mitigation measures and implementation of these measures through an environmental management plan.

1.5 Physical Infrastructure Development Activities in SSRP

One of the main activities of the SSRP is physical infrastructure development of schools and District Education Offices. The EMF is directly related with the physical infrastructure activities of SSRP. Under the physical infrastructure component of SSRP, the following activities are envisioned;

1. Construction of administration buildings (Construction of District Education Office building, total = 20 nos.)
2. Construction of school and hostel building for disables (Total = 100 nos)
3. Construction of library and laboratory (Total = 300 nos)
4. New classroom construction (Total = 19500 nos). Out of this, 75% will be built under the regular modality and 25% by additional contribution (cost sharing basis)
5. External environmental improvement (construction of girls toilet, boys toilet, water supply)
6. Classroom rehabilitation (Total = 13000 nos).

The implementation of EFA 2004-09 has been a remarkable experience and achievement' for the Ministry of Education (MOE) of Nepal. Encouraged by these achievements, the Government has decided to enter into a new phase of program implementation. The SSR is a response to the aspirations of the people and to emerging challenges the education system of Nepal is facing. The SSR should be viewed as a combination of affirmative actions and policy changes towards attaining the EFA goals by 2015, and be seen as a commitment towards ensuring quality of education to meet the local and global needs.

Table 1.1: Implementation Mechanism for Physical Infrastructure Development

S.N.	Activity	Implementation Mechanism	Responsibility	Timeframe	Indicators
1.	Classroom Construction	<u>School Selection</u> • <u>Physical Survey of Schools</u> Technical staff of DEO will use the standard format developed by EOE & conduct a survey of overall school in the district • <u>Preparation of School Physical Facilities Improvement Plan (SPIP)</u> SMC will prepare SPIP. This includes activities to be performed in schools, estimated cost to perform such activities and contribution of schools in	DOE/PSS DEO		School selection criteria is given in annex I

S.N.	Activity	Implementation Mechanism	Responsibility	Timeframe	Indicators
		such activities • <u>Analysis of Education Data</u> Insert date on Education Management Information System (EMIS) & analyze educational indicators	SMC		
		<u>Prioritization of Schools Selection</u> Based on educational indicators given in Annex – I school are prioritized for support <u>Program Setting</u> The whole school approach concept shall be followed in order to completely include all the facilities for school & get approved from DEO. The approval of DEO/AREA officer is required prior the expenses of such construction budget by the school. <u>Preparation of Cost Estimates</u> Prepare cost estimates of each individual components of school physical facilities and getting approval from DEO. The district rates & GoN norms should be followed in cost estimates. The cost should include the contribution from school & program. <u>Agreement with School</u> Based on the cost estimate & indicating the clear cut responsibilities of local community an agreement is signed between DEO & SMC. The name of schools & list of activities/programs to be conducted in such schools, DEO will arrange technicians <u>Release of Funds to Schools</u> Upon technical clearance for construction of such selected schools, the fund will be released to respective schools. <u>Expenditure by School is tied with work Progress</u>	• DEO/PSS • DEO		

S.N.	Activity	Implementation Mechanism	Responsibility	Timeframe	Indicators
		Based on progress of work, the fund will be released/spent by the school. <u>Documentation</u> DEO shall prepare a documentation related to construction of school (i.e. Physical survey form, agreement, evaluation document, measurement book, report). Each 2 sets of documents are prepared. One set will be retained at school & another set shall be forwarded to DEO. <u>Supervision of Construction Works</u> DEOs will assign technical personnel (Engineer/Sub Engineer) to supervise & monitor construction activities in schools. For this, there is also a practice of utilizing technicians through NGOs/INGOs as per the MOU. <u>Construction should Comply with Agreement</u> Construction should comply with clauses of agreement. The breach of contract, sub standard work & delay could subject to legal action by DEO to chairman of SMC & personnel involved. • The cost allocated for new classroom construction is inclusive of 5 sets of furniture. • The new classroom should have black board of size (8'X4'). • There should be provision for RAMP to wheel chair users.			
2.	Support to School Water Supply System as a part of improving external environment	1. Mechanisms same as 1 2. Other Requirements a. Water Availability for drinking & sanitation. b. Tube well should be at least 15m from soak pit or septic tank. c. The fencing of school & external environment improvement is entirely the responsibility of school.	• DOE • DEO • SMC • Community	II & III Trimester	
3.	Separate Toilet for Boys &	1. implementation Mechanism same as 1 2. Other Requirements a. The boys & girls toilet should	• DOE/PSS • DEO • SMC		

S.N.	Activity	Implementation Mechanism	Responsibility	Timeframe	Indicators
	Girls	maintain a distance b. Construction of RAMP for wheel chair users c. Availability of water in school for water supply & sanitation purposes d. Other requirements as of no. 2	Community		
4.	Construction of District Education Office Building (Including furniture)	- Based on DOEs design/drawing the cost estimates shall be prepared. The norms shall follow district & GoN. The procurement of work shall follow as per procurement guidelines of GoN. The tender document shall be prepared. 1 set of tender document should be sent to DoE/PSS. - Publish tender notice - Evaluate the bids - Agreement with bidders. A copy shall be forwarded to DoE/PSS - Under the close supervision of technician construction work should be done	DoE/PSS DEO	I, II, III Trimester	

Chapter II - Environmental Assessment

2.1 Methodology Used to Prepare EMF

The present EMF was prepared based on the following methodologies:

1. Review of existing literatures including the following:

- a. Plan, policies, and programs of MOE/DOE related to the improvement of school physical facilities
- b. Environmental Protection Act (1997), Environmental Protection Regulations (1997)
- c. National plan, policies in education sector in Nepal, Self Governance Act, Educational Act and Regulations
- d. EMIS format and data collection procedure
- e. Study reports (basic design, EFA reports including core document and Environmental Assessment, SESP core documents-2008)
- f. Implementation status review report of EMF and other safeguard documents.
- g. Alternative design of classrooms, guidelines for the development of school facilities under EFA including standard, type-designs and technologies used in different geographical regions and climatic conditions in Nepal.
- h. National Environmental Guidelines for School Improvement and Facility Management in Nepal, 2004.
- i. Other reports and documents as found suitable by the DOE for the preparation of above framework
- j. Practices under BPEP II and EFA with respect to the construction and rehabilitation of classrooms.
- k. Practices under EFA related to the school physical facilities improvement including but not limited to drinking water supply and sanitation, play grounds, compound protection, drainage, solid waste disposal, landscaping with greenery
- k. Current practices of Operation and Maintenance (O& M) of buildings, water supply, drainage, sanitary facilities, sports facilities, school compound and other utilities including funding for O & M of facilities
- l. The environmental safeguard policies of SSRP development partners
- m. Consultations were carried out with central, district and local stakeholders with regard to environmental considerations to be included in SSRP.
- n. Field visit for getting the perception of stakeholders and observe “on the ground” activities accomplished in EFA with respect to implementation status of Environmental Guidelines prepared under EFA with particular reference to its dissemination, training of personnel, institutional set up, monitoring framework, and revisions of designs and construction practices. Based on the analysis, specific and realistic actions will be recommended for improvement while implementing the SSR. The recommendations are accompanied by appropriate and practical formats that can be used during implementation and monitoring of SSRP. The lessons learned, lacunae and useful measures that could be replicated in SSR shall be assessed and used invariably in Environmental Management Framework for SSR.
 - The chronology of consultative meetings held were given in annex II

- The discussions were also held with experts and the concerned staff of DOE, relevant organizations working in education sector in Nepal. The list of persons/institutions contacted during the preparation of EMF is given in **annex III**
- The observation from the field visits are listed in **annex IV**
- Summary of stakeholders feedbacks on environmental issues related to SSRP are listed in **annex V**.

2.2 Policy, Legal, and Institutional Framework for Environmental Management

In this section the policies, legal and institutional framework for environmental management in Nepal are summarized including national legislations.

2.2.1 Review of Nepal's National Environmental Policies

National Policies: Development planning exercise, which was started in mid 1950 in Nepal focused on natural resources utilization agriculture production, sanitation, and sewerage facilities and infrastructure development. Only in 1970s the need for the integration of environment aspects in development planning was realized. Since then efforts are being made to ensure that environmental policies and programs will be an integral part of national planning.

The Sixth Plan (1980-85) adopted policies and programs on land use, soil conservation, and watershed management, development and management of national parks and wildlife reserves. The policy focus was on the implementation of the integrated development programs, environmental studies of development projects and control of air, and water pollution in the early 1980s. Exports of forest products were banned in 1980s.

Subsequently, environment was recognized in Seventh Plan (1985-1990) as an important aspect to be integrated into every development projects. Some important measures were undertaken for environmental protection and management. Importance of public participation in the decision making process, and the role of women and non-governmental organizations were also emphasized for the management of environment. The Eighth Plan (1992-97) re-enforced to sustainable economic growth, and poverty alleviation. The Plan emphasizes the need for internalizing the Environmental Impact Assessment system particularly in economic development plans and programs. A National Environmental Policy and Action Plan (NEPAP) were prepared in 1993 to facilitate the integration of the environment aspects in development process. His Majesty's Government of Nepal's (HMG) environmental policy was also clearly spelled out in NEPAP. There are five main aims of HMG's environmental policy:

- To manage physical resources in an efficient and sustainable manner
- To balance development efforts and environmental conservation for sustainable fulfillment of the basic needs of the people
- To safeguard national heritage
- To mitigate the adverse environmental impacts of development projects and human actions
- To integrate environment and development through appropriate institutions, adequate legislations and economic instruments, and sufficient public resources.

During this period, master plans for irrigation and livestock were also prepared. Environmental aspects in the policies of hydropower, irrigation and industry were incorporated. The Agriculture Prospective Plan was prepared and implementation started. The Ninth Plan (1997-2002) was formulated to identify system for ensuring maximum participation of general public in environmental decision making process, fixation of environmental standards, establishments of strong environmental

information and management system, implementation of international treaties, engagement of NGOs and INGOs in environmental sector, initiation of environmental health programs etc.

According to the Eighth Plan, causes for air, water and land related pollution was supposed to be investigated through on-the spot observation and management mitigation plans. In this regard emphases were laid on adopting technology required to minimizing waste and neutralizing it. However pollution control program launched during this period included limited assessment studies of existing situations in the areas of solid waste along with other sectors: water, air, noise, etc. Utilization of appropriate technologies still needs to be developed in the areas of solid wastes along with other areas. Research on the cause of air, soil, and water related problems still need to be investigated and analyzed to prepare indicators and action oriented guidelines for the utilization.

The solid waste aspect has been mentioned in the Ninth Plan. The Plan includes: consideration for the engagement of NGOs and private sector for the management of solid waste, emphasis for composting for municipal wastes, setting norms and standards in this field and its strict implementation in every municipalities and capacity building of local municipalities for handling solid waste related issues.

Though the development plans recognized the adverse environmental deterioration and importance of various remedial measures, the policy lacks the most needed coordination, flexibility, and responsiveness as regard to the solid waste related issues.

Similarly in the Tenth Plan objectives are set up for environment conservation that is indirectly related to school environment and addressed:

- To take into account that sustainable development exists through the inter relationship between environment conservation and economic development; emphasize conservation, security, re-establishment and sustainable utilization of natural environment.
- Promoting sustainable development by controlling pollution and amassing public participation in a campaign to create a clean and beautiful rural environment.

The main strategies in the area of environmental management in Tenth Plan are (i) internalization of environmental concerns into development projects and programs (ii) fixation and implementation of national environmental standards (iii) forging partnership with local institutions, NGOs and private sector in awareness campaign (iv) preparation of action plan and implementation of international environmental treaties and conventions. The main policies are (i) capacity development of local institutions in environmental management (ii) promotion of women participation in environmental management at national as well as at local level (iii) encourage scientific research and development to develop environmentally friendly technologies and (iv) develop appropriate legal and fiscal mechanism to manage and control industrial pollution. The main program is Environmental Conservation Program.

2.2.2 Review of Environmental Laws

2.2.2.1 The Constitution of Kingdom of Nepal 1990

Article 26(4) embodies following policy mandates: “The state shall give priority to protection of the environment and also to the prevention of its further damage due to physical development activities by increasing the awareness of the general public about environmental cleanliness and the state shall make arrangements for the special protection of the rare wildlife, the forest and the vegetation.”

With the above provisions, some constitutional responsibilities and duties have been vested upon the State for safeguarding the environment against its further damage due to physical development activities. (Pant, 1998)

2.2.2.2 The Environmental Protection Act 1997

In order to give effect to constitutional mandates special “Environmental Protection Act 1997” (enforced from June 24, 1997) and “Environmental Protection Rules 1997 (enforced from June 26, 1996) have brought into existence. These Act and Rules have made provision dealing with pollution control, EIA, IEE, conservation of national heritage etc.

Section 24 empowers the GON to frame rules including for matters relating to sources of pollution, standards, prevention and control of pollution. There are also several sectoral Acts related to environment.

2.2.3 Other Related Legislations & Guidelines of the Government of Nepal

2.2.3.1 Local Self Governance Act (1999), Local Self Governance Rule (2000): The Local Self Governance Act has been enacted to provide greater political, administrative and financial autonomy to local bodies and facilitate community participation at the local bodies and community participation at local level. In accordance with the Act, local bodies have been formed at three levels: Village of Development Committee (VDC) at village level, Municipality at town level, District Development Committee (DDC) at district level. The Act empowers these local bodies to formulate and implement periodical and annual plans within their own jurisdiction.

2.2.3.2 Forest Act (1993), Forest Regulations (1995): The use of forest land is subject to forest act and regulations. The school projects need to comply with the provisions of forest law when it requires the use of forest land.

2.2.3.3 Public Work Directives, 2002. The directives provide general guidance pertaining to public works including construction of buildings. These include procedures and requirements related to: environmental assessment; social assessment; earthquake-related considerations in infrastructure projects; occupational health and safety; environmental management in civil/construction works; and specific sectoral procedures to be followed.

2.2.4 Policy for the Improvement of School Facilities

The Government of Nepal’s effort towards school physical facilities development through various programs till date is presented in **Annex –VI**. The DOE has joined hands with INGO/NGO in quality supervision for the construction of school physical facilities. Out of 75 districts, in 32 districts INGO/NGO is active in providing quality supervision of the construction of school physical facilities. Among those are the Save the children Norway are working in 20 districts, Plan International in 6 districts, Room to read in 3 districts, Save the Children (US) in 2 districts, and Save the Children Japan in 1 district. The DOE policy for the design and construction of school buildings are presented in **Annex-VII**.

2.3 Summary of Provisions that are Applicable to SSRP Activities

The environmental issues in SSRP were not anticipated to be grave. No large scale, significant/or irreversible impact is anticipated. The environmental review has identified several small scales & local environmental potentially associated with the program activities.

According to Environmental Protection Regulation (EPR 97), there are two levels of environmental assessment, namely, IEE & EIA. Schedule 1 of EPR lists projects that require IEE & schedule 2 lists projects that require EIA. Their criteria are based on the nature of size of the project & its cost. In

addition, the EPR also list environmentally sensitive areas, where any proposal, regardless of the size & cost will require & EIA.

Since the nature of activities under SSRP doesn't trigger any national or international environmental screening requirements, a workable environmental manager plan has been proposed in EMF for SSRP. While preparing the SIP, all the environmental issues related with improvement program will be identified so that these issues will be internalize in the planning, design, and construction as per the EMF which will be followed by preparation of EMPs. However, sub-projects activities like construction of annex buildings, laboratory and toilets etc. will undergo a screening to determine the type and severity of the impacts and EMPs will be accordingly prepared if required. For identifying the key issues, environmental screening has been prepared **Annex – VIII** which can be customized for both aforementioned tasks Based on the screening, a workable environment management/mitigation activity has been tailored.

2.4 Review of Safeguard Measures

The following paragraph analyses safeguard measures assessed by different development partners. The assessment also reveals status of the safeguard in relation SSR Plan implementation.

S.N.	Development Partners	Safeguard Policies
1.	International Development Association (IDA)	EA policy OP 4.07 is triggered. According to this SSRP has been categorized as Environmental Category B Project . Thus in case of SSRP, environmental impacts due to program activities and likely to include loss of vegetation, soil erosion, water pollution, air pollution, health impacts are children, proper sanitation measures must be fully identified & appropriate mitigation measures clearly defined & cost to be incorporated into program's overall budget.
2.	Aus AID	The SSRP falls under scenario 1 of environmental marker set by Aus AID, which states <i>no significant environmental impact</i> . The monitoring of Unexpected environmental impacts should be incorporated into activity's monitoring & evaluation process.
3.	EU	The liability for environmental damage falls into three categories namely; protected species & natural habitats, water & soil. The SSRP activities won't trigger any negative implications in above three categories.
4.	ADB	The SSRP falls under category C (unlikely to have significant environmental impacts). No environmental assessment is required for category C projects although their environmental implications nevertheless need to be reviewed.
5	UNICEF	Helping schools provide supplies, safe water and sanitation. Water, sanitation and hygiene are crucial to getting and keeping girls in school, as they bear the brunt of unhygienic or non-existent latrines. The lack of clean and separate sanitation facilities in schools discourages many girls from attending school full time and forces some of them to drop out altogether, particularly as they approach adolescence and the onset of menstruation. Lack of water in the household also keeps girls away, as they are usually the ones designated to walk long distances to fetch the household's water supplies. And children of both sexes are sapped of nutrients, energy and the ability to learn if they are infected with water-borne parasites. This issue has been well addressed in EMF of SSRP.
6	FINLAND	Emphasize the importance of environmental considerations in all activities with a view to forestalling threats, and creating conditions conducive to prosperity based on the sustainable use of natural resources; • support environmental programmes and the development of environmental administration; • take into account the principles of sustainable development in supporting agricultural and forestry programmes;

S.N.	Development Partners	Safeguard Policies
		• support developing countries in fulfilling their environmental obligations; and • Practice and further develop procedures emphasizing the independent decision-making authority and participation of the partner in cooperation and relevant interest groups.

The SSRP is not likely to present any perceived environmental risks and potential adverse impacts in its area of influence. Hence, the nature of activities under SSRP doesn't trigger any national or DP's specific environmental assessment requirements.

2.5 Environmental Issues and Its Significance for Infrastructure

Development under SSRP

Environmental Assessment (EA) is a process whose breadth, depth, and type of analysis depend on the nature, scale and potential environmental impacts of the proposed program. For SSRP, EA takes in account the environmental issues highlighted above below.

The SSRP is not likely to present risks and potential adverse environmental impacts in its area of influence. However, the associated risks posed by seismic hazards should be given due consideration during planning of SSRP activities as well as to the risks of floods, landslides, fire etc. Thus in the case of SSRP, the environmental and social issues addressed below have been categorized and the appropriate mitigation measures and its implementation arrangements, capacity building programs associated with it are clearly elaborated in EMF. Multi hazard risks should be addressed appropriately.

This EMF has been designed to address such environmental and social issues at the planning, construction and operational stages under the physical infrastructure development activities of SSRP

S.N	Environmental Issues	Significance	Remarks
1.	Slope stability and land slides	Moderate	Schools in mountain and hill region of Nepal fall on slopes. Poor site selection for school building construction lead to instability, and lead high cost on account of earth retaining structures.
2.	Location of school and orientation	Moderate	This is very important as it determines magnitude, natural heating, and lighting available inside the building. Orientations should be based on the climate type of the place where school facilities are planned
3.	Sanitation	Moderate	Adequate sanitation facility in school (toilets, personal hygiene practices) should be encouraged.
4.	Water Quantity/Quality	Moderate	An easily accessible water supply that provides sufficient safe water to meet school needs. Schools in Nepal are found using sources like protected springs, dug wells, boreholes, and piped water supply. The problem of arsenic contamination in ground water in schools of Terai should be

S.N	Environmental Issues	Significance	Remarks
			given special attention.
5.	Wastes	Low to moderate	Solid waste should be disposed of properly. Key strategies for improving solid waste management and disposal are to minimize the waste by schools and to recycle waste whenever possible.
6.	Day Lighting and windows	Moderate	Poor indoor lighting in classroom can have many harmful effects on health and well being of teachers and students.
7.	Pollution (Air, Water, and Noise)	Low to moderate	The air, water and noise pollution issues during the construction and operation phases of the school and administration building construction should be addressed. The air pollution caused by old school buses is a subject of concern and need attention for regular repair and replacement. The air pollution caused by vehicular movement in school areas is another issue to be considered. Proper speed control measures and air pollution control measures shall be incorporated. In many schools, teaching-learning activity in one class room disturbs the other class room(s) (because class room designs are inappropriate from noise consideration).
8.	Ventilation systems	Moderate	The poor ventilation in schools may lead to respiratory problems, and easier transmission of infectious diseases. Constructing schools with sufficient number of windows may improve ventilation.
9.	Seismic Hazard	High	Earthquake safe school and administrative building should be encouraged.
10.	Natural Hazards (storm, wind, Thunder, Hail stone etc)	Moderate to High	Maximum care should be taken during design of structures and emergency planning measures should be integrated.
11.	Drainage and flood management	Moderate to High	Drainage management should be an integral part of construction of schools. Construction of schools in flood plain areas shall be given special attention. Necessary control, emergency measures should be an integral part of the program.
12.	Health and Hygiene	Moderate	The health and hygiene education and practices offered in schools was felt to be inadequate.
13.	Construction materials use	Moderate	The use of local materials should be encouraged. Avoid use of wood in construction of schools in Terai. Special attention should be given to control deforestation.

S.N	Environmental Issues	Significance	Remarks
			The quarrying of sand from River should be limited to avoid any negative environmental impacts. Local quarrying of stone and other raw materials should be within limit not jeopardizing the local environment. Design and implementation of climate friendly school building construction.
14.	Laboratory safety and hazardous waste	Moderate	The laboratory safety and hazardous waste management in majority school laboratories in Nepal is poor. So, it is advisable that all recipient schools for the laboratory under SSRP should the checklist included in annex of this EMF.
15.	Safeguarding the school facilities	Moderate	To define school land, to safeguard students, school buildings and resources, to prevent entry of animals to limit the public access to public access to school and for various safety and security purpose, fencing of school is essential.
16.	Class room temperature	Moderate	In hot climate, e.g. in Tarai plain area, using GI sheet roof in school means subjecting school kids to high temperatures. Innovative ways need to be looked at to address this issue.

2.6 Past Experiences

2.6.1 .Components of NEGSIFMN that Worked

The school building design and construction standard developed by the DOE has incorporated the standards and good practices prescribed in NEGSIFMN. This standard has clearly spelled out that all the construction should follow the National Building Code, and child friendly constructions. This has been reflected in their guiding principal, which states “Child friendly (well lighted and well ventilated), seismically resistant, aesthetically pleasing school maximizing the use of local construction material and resources with no adverse effect on environment like deforestation, and having low maintenance requirements.”

In program implementation guideline prepared by the DOE (2007-2008), the following aspects of NEGSIFMN have been incorporated in component 1 annex of the above manual.

- The school must meet the drinking water quantity/quality standards.
- The distance of 15 meter must be maintained between the septic tank and tube well.
- The school boundary/fencing wall should be constructed by the school from their own source to make the school safe.
- Provision of Rs 1 lakh has been allocated for the construction of separate toilet for boys and girls
- The guideline has discouraged the use of “timber” in Terai and promotion of steel tubular truss.
- The provision of ramp in classrooms and toilet for disable students.

The following aspects were improved during the period through the intervention of NEGSIFMN.

- The awareness has been raised regarding the arsenic contamination in wells of Terai Schools and safe drinking water.
- Promotion of hygiene awareness programs in schools
- Separate construction of girls toilet
- The partnership with INGO/NGO for the effective monitoring in school physical facilities construction.
- The environmentally friendly design guidelines for the construction of school under EFA are very useful. In SSRP, this aspect is also recommended for inclusion through EMF see **Annex IX**.

2.6.2 Components of NEGSIFMN that did not Work

There are several components of NEGSIFMN that were less successful during EFA. However, even failures can teach valuable lessons. Some of the aspects that didn't work in NEGSIFMN are explained below.

Integration of NEGSIFMN in planning process i.e. School Improvement Plan (SIP) was missing. The capacity building activities for the implementation of NEGSIFMN was spelled out in the document but failed to incorporate it in the project cycle in EFA. The non allocation of budget and program for the implementation of NEGSIFM were some of the shortcomings. Due to this reason, the proper dissemination of NEGSIFMN and its implementation couldn't take place in EFA.

The research and development aspect in alternative of school design, selection of construction material was also found missing.

2.6.3 Innovative Approaches

Some of the innovative approaches so far being adopted by DOE are highlighted below.

- Alternative designs prepared with the use of local materials (22 type model, compressed stabilized earth blocks (CSEB) model under approval).
- The community must get an approval from the Department of Urban Development and Building Construction (DUDBC) for their own designs.
- Trainings were provided to engineers and sub engineers for earthquake safe design and construction.
- Provided orientation to SMC, teachers and masons on construction quality.
- Construction of physical facilities through community participation (at least 25% community participation)
- Design of RCC school building is available and school willing to construct such RCC building (through cost sharing provision) is selected. For the construction of such building DOE shall allocate Rs. 825000.00 per scheme.
- DOE has a provision for the construction of separate toilet for boys and girls (Rs. 100,000.00 per school) has been allocated under this scheme.
- In order to improve the water supply system in schools, DOE has allocated Rs. 10000.00 for the improvement of school water supply systems
- Bardiya District Education Office in collaboration with MS Nepal, and BASE Bardiya built tow pilot models of the new design in Bardiya districts. The blocks/bricks are produced using Compressed Stabilized Earth Block (CSEB). The design is based on climate responsive (also called bio-climate) principles to nationally maintain a relatively comfortable temperature in all seasons. The research and development is ongoing regarding this technology.
- DOE has pilot tested Carbon Fiber UPVC roofing in schools. This roofing material is easy to fix, inflammable, acid/alkali erode proof, sound and heat insulation properties. The detail technical and financial feasibility is yet to be done by DOE prior its use in schools.

- In order to establish good water quality system in schools, DOE has collaborated with Department of Water Supply and Sewerage and UNICEF in improving water supply systems in schools
- As agreed with DOE some INGO/NGOs have provided technical support for school building construction in 32 districts.

2.6.4 Lessons Learned

2.6.4.1 Performance Review

The performance status review of EMF document (drafted in 2009) and its compliance for SSRP was conducted through random sampling of schools funded through SSRP. The findings indicated that the provisions, process and procedures in the existing EMF are adequate to address all the pertaining environmental issues which arise during the construction activities. On the other hand, the main challenge is associated with insufficient program monitoring from DOE/DEO, partly due to inadequate technical human resources, budget and institutional arrangements. The environmental screening, the first step to identifying the potential concerns as spelled out in the EMF, was found to be missing in a few subproject activities of schools.

NEGSIFMN

The development of NEGSIFMN was an innovative approach.

Clearly defining and specifying the coverage, contents, and applicable standards of such components in EMF would eliminate the issues in SSRP. The extended and revised version of NEGSIFMN would definitely overcome such lacunas in meeting the overall objectives of SSRP. Considering these, the positive aspects of NEGSIFMN should be maximized and the shortcomings in EFA should be overcome through the application of the EMF under SSRP. These aspects are dealt with under the capacity building for implementing the EMF in SSRP.

2.7 MECHANISMS FOR IMPLEMENTATION OF EMF IN SSRP

S.N.	Activities	Responsibilities	Remarks
1.	Preparation of EMF, Its Approval, Circulation & Incorporation in Program		
1.1	Approval of EMF for SSRP	MoE	
1.2	Printing, Publication of EMF in Nepali & its wider circulation	DoE	DOE provides EMF guidelines to DEOs for implementation
1.3	Dissemination of EMF through Program Implementation Manual (PIM)	DoE/DEO	DoE will ensure that stakeholders e.g. DEO, SMC and others understands the EMF requirements and EMF provisions are followed while planning and implementation of School Physical Improvement programs (SPIP) and DEO Office buildings.
1.4	Incorporation of EMF aspects in SIP	DEO/ SMC	DoE will ensure that EMF is an integral part of decision making for SSRP activities.
1.5	Dissemination of EMF through regular orientation programs organized for Engineers, Sub-Engineers, District Education Officers, I/NGOs and other partners working in collaboration with DoE.	DoE/ DEO/ Partner INGOs	Engineers, Sub-Engineers and Partner INGO's representatives are involved in exercises on how to incorporate EMF provisions in planning and implementing SIP
2	Environmental Assessment & Management		
2.1	No special EA is required as construction activities are of very small threshold, i.e. less		

S.N.	Activities	Responsibilities	Remarks
	than NRs 825,000.00		
2.2	Environmental Screening of sub-project activities	DEO	Prepare EMP if required
2.3	Incorporate EMF requirements in SIP preparation guidelines	DOE	Follow approved EMP
2.4	Utilize EMP requirements in school construction, as spelled out in SIP	SMC	
2.5	For DEO Building Construction in Corporation of EMP Aspects in contract document	DEO	Clauses in tender document should be clearly defined
3.	Monitoring & Auditing		
	a. Field Level Monitoring – During implementation – Four monthly, feeding into government's quarterly portfolio review meeting – Site visited to representation samples	DEO (as an integral part of ASIP monitoring system) a dedicated technical person preferable ,	Checks screening report & incorporation of EMPs recommendations into subproject, field implementation, & record keeping also check progress report
	b. Sub Project Level Monitoring – During pre-construction, construction & operation stages – Regularly	By DEO at regular basis & DoE by intermittent & as cross check Prepare status report challenges, obstacles faced	Keeps updated records of the mitigation works undertaken Report to DoE/PSS
	c. Compliance Audit – During Implementation – Annually	DOE, Central Level Team (As an integral part of overall program implementation audit) Visits random representative	Verifies overall compliance to EMF & other env. & social safeguard requirements
4	Annul Reporting & Feedback – Interactions/Workshops – Reports	DoE (DoE) reports status on the ground reality, & recommendations for improvement to MoE)	DoE will conduct an annual interaction to replicate the best practices & to avoid/ mitigate obstacles problem

2.8 Existing Capacity Assessment for the Implementation of EMF

S.N.	Organization/ Agency	Existing Capacity			
		Knowledge/Experience of Env. Mgmt	Manpower	Budget	Logistics
1	MOE	X	X	X	X
2	DOE	√	Engineer assigned additional environmental tasks, over and above	X	X

S.N.	Organization/ Agency	Existing Capacity			
		Knowledge/Expe rience of Env. Mgmt	Manpower	Budget	Logistics
			engineering tasks.		
3	DEO	X	Engineer assigned additional environmental tasks, over and above engineering tasks.	X	X
4	SMC	X	X	X	X

Note: X: Lack of capacity, √: Adequate capacity

The physical services section of the DOE is taking responsibility for the physical infrastructure related activities since BPEP II. This responsibility of DOE has been continued for SSRP as well.

DOE based on pre set district targeting criteria allocates program and budget to all 75 DEOs. DEOs based on pre set school selection criteria allocate program and budget to recipient school. Schools are fully responsible to implement the program under the technical supervision of DEO.

The DOE, DEO has very limited technical staff, hardly sufficient to run their annual programs. There is no environmental specialist at any tiers (neither at MOE, nor at DOE or DEO) . An environmental specialist was hired earlier by DOE(Physical service section) to provide environmental oversight and for coordination of environmental activities for all infrastructure works executed by DOE. But as of now, there is a civil engineer who has been assigned for this job and now needs to provide overall environmental oversight, coordination and help implement the EMF for SSRP. The officer should also be responsible in mainstreaming the environmental best practices, emerging lessons & innovative ideas, and coordinate/facilitate environmental management through various schools and DEOs in the country.

Similarly, the SMC and schools also do not have access to environmental manpower and knowledge to deal with issues related to environment mentioned in NEGSIFMIN and EMF of SSRP. Their capacity in terms of manpower and resources is required in order to implement EMF in SSRP. Therefore, provision for dedicated technical person, preferably an environmental specialist in the DEO or in the team of service providers needs to be made for monitoring the construction activities at the local level..

Chapter III - Environmental Management Framework (EMF)

3.1 An Overview of Environmental Steps in SSRP

The EMF has been developed by the Ministry of Education, Nepal for the SSRP in order to ensure due diligence and to avoid any environmental degradation issues. The purpose of this framework includes assisting the SSRP implementing agency DOE in dealing with issues related to environmental impacts, identifying suitable mitigation measures and implementation of these measures through an environmental management plan.

3.1.1 Environmental Assessment (EAs)

Considering the nature and magnitude of potential environmental impacts from relatively limited small scale construction, renovation, refurbishing construction work and other associated components of SSRP, the proposed operation is to be classified as category “B”.

The SSRP is not likely to present any perceived environmental risks and potential adverse impacts in its area of influence. Hence, the nature of activities under SSRP doesn't trigger any national or international extensive environmental requirements. The SSRP, as a whole, doesn't require any environmental clearance. However, in order to manage the minor and localized impacts GON has suggested developing a screening and preparation of EMP for schools. The proposed EA suggests the following steps;

3.1.2. Integration of EMF in SIP

The provision made in EMF will be well integrated in the SIP to address the key environmental issues. The integration of EMF in the School Master Plan/School Improvement Plan (SIP) will be one of the criteria for selecting the school by DEO. EMF provides an essential link between the impacts predicted and mitigation measures specified. EMPs/mitigation measures will be included in the design, contract and bidding documents for implementation. Sample design Guidelines for constructing environmentally friendly school building from Environmental Perspective is given in Annex - 9.

The EMP should include the following: i) a site map showing environmental features of the school and its surroundings, ii) concise description of the environmental risks/ concerns relevant to the school, and proposed SIP as well as potential solution to the risks/ problems identified (summarize these in a table).

3.1.3 Environmental screening for sub projects

For any sub project activities which will involve infrastructure construction, renovation/refurbishing, construction works, district level offices and institutions meeting the eligibility criteria will use a checklist (Annex - 8) for screening to identify issues and environmental impacts related with sub-projects.

The screening process will be followed by drafting mitigation measures or preparation of EMPs if required accordingly.

3.2 Mitigation

The EMP identifies site-specific, cost-effective and detailed measures that will reduce the identified adverse environmental impacts to acceptable levels. The plan should include compensatory measures

if mitigation measures are not feasible, cost-effective, or sufficient. Examples of the environmental mitigation measures that may be appropriate to SSRP are shown in table below. The specific EMP must have its own mitigation measures that are appropriate to the issues/ risks/ difficulties identified through the process of screening and EMP preparation.

Environmental Mitigation Measures

Category	Potential Problem	Appropriate Mitigation Measures
Slope stability	Landslide or other forms of mass instability on the slopes	Geological/geomorphologic considerations shall be incorporated in design Civil engineering structures and bio-engineering measures used. Measures taken to avoid undercutting of slope toes. Quarrying prohibited in river beds, where flood discharge is significant
	Development of erosion or gulying in school premises	Check dams and bio-engineering measures used as necessary
Spoil disposal	High volume of water/spoil	Minimize spoil by balancing cut and fill wherever possible in school building construction
Water Management	Water logging, poor drainage	Proper drainage management, Improve and increase slope drainage measures
Water Quality/Quantity	Sub standard or inadequate drinking water provided to students	Ensure that enough drinking water is supplied to school. School shall request the drinking water supply system construction/repair budget from the program. Regular testing of drinking water at least 2 times a year is recommended In schools of Terai, arsenic detection in ground water is recommended. The arsenic monitoring and mitigation plan suggested in EFA shall be continued
Land acquisition and resettlement	Land transfer will be completed through registration, with land title vested in the School, Leasing of land or rights of use of land for schools will be duly documented	Resettlement and or land acquisition problems are not anticipated in SSRP. The DoE will not do any involuntary land acquisition. All donations and purchases will be voluntary. DoE will not only ensure that any land donations are fully voluntary, but also that such donations do not involve physical displacement or any significant adverse impacts upon incomes of the donor household. All voluntary land transactions will meet the criteria as mentioned in annex VI.
Plants	Trees being removed in school premises	Felled trees replaced, using the same species if appropriate. Trees planted wherever land is available.
Quarries and borrow pits for obtaining construction	Pollution, disturbance and danger from quarry operations	Construct bunds to screen noise and dust. Enforce access restrictions.
	Safety risks from abandoned	Quarries made safe by re-grading slopes and installing

Category	Potential Problem	Appropriate Mitigation Measures
materials	quarries or borrow pits	structures as necessary.
	Land seriously disturbed or lost from production	Quarry and borrow areas rehabilitated to productive plantations using bio-engineering techniques.
	Quarries continue to be used by unauthorized persons	Unauthorized quarrying stopped, where necessary by working with the District Development Committee.
Stone crushing	Dust and noise pollution	Spray water in crusher machine, Use it in off hours of schools. The facilities should be away from school premises.
Air and noise pollution	Dust generated from construction works, and approach roads to school	If water is available, the road surface can be sprayed on a frequent schedule.
	Air pollution generated due to old school buses Dust and noise generated by vehicles passing by schools	Old school buses should be regularly phased out. Regular repair and maintenance of vehicles. Traffic sign should be kept mentioning “School Zone”, Speed control measures bumps etc should be kept
	Noise during construction of schools	Work schedule to minimize disturbance.
Social issues		
	Local people excluded from project activities	Designs incorporate methods within the skills of local people. Contractors encouraged using local labor wherever possible.
	Promises were made to local people during feasibility and planning phases	Checks to ensure that the promises are fulfilled; if they prove to be not possible, reasonable alternatives must be negotiated.
Solid waste management	Spreading of waste, pungent smell, deterioration of aesthetics	Proper solid waste management system should be introduced in schools which includes segregation of waste, and its proper disposal. The environmentally friendly management measures like composting should be encouraged. Awareness regarding the management of solid waste, discouraging use of plastic products etc should be promoted in schools The use of recycled materials should be promoted.
Hazardous waste	Use batteries, laboratory chemicals etc	Safe disposal of hazardous waste should be encouraged.
Health and hygiene, and food safety	Unhealthy living condition in schools, disease	Promote health measures in schools. (hand washing facilities, safe disposal of human excreta, availability of water, etc). The norms prescribed for toilet by NEGSIFMIN should be followed.
Water	Lack of quantity of water, poor	Water conservation measures should be incorporated in

Category	Potential Problem	Appropriate Mitigation Measures
Conservation	sanitation	schools such as (efficient and rational use of water, incorporation of rain water harvesting systems in schools)
Energy Efficiency	Use of non energy efficient bulbs, and equipments in schools	Use energy efficient systems in schools, the natural lighting and orientation of school building should be integral part of design. Use of renewable energy like solar should be promoted in schools
School safety	Theft, burglary, damage of physical facilities	Adequate fencing of schools is required. The windows of classrooms should be secured by “grill” for any kind of trespassing by students and outsiders.
Construction materials Use of Wood Material specifications	Deforestation	The use of wood for the construction of school physical facilities in Terai has been banned. The innovativeness to avoid use in hills and mountains should be integrated in schools. The recycled materials to reduce use of raw materials. The use of local materials as much as possible The new innovations should be integrated in design for making schools more child and environmentally friendly.
School safety (Fire, earthquake, hail stone, thunder storm etc)	Damages to school properties and possible human casualties of different scale	Incorporate safety measures during design of physical facilities Create awareness about natural calamities to teachers and students

3.3 Implementation of Mitigation Measures

a) Program Design: The mitigation measures should be integrated in the design of the program itself. Such a step will enhance the mitigation measures in terms of specific mitigation measures in terms of specific mitigation design, cost estimation of the mitigation measure, and specific implementation criteria. The mitigation measures integration in the design phase will also help in strengthening the benefits and sustainability of the program. **b) Implementation Modality:** Most of physical construction works (new classroom construction, rehabilitation of old classrooms etc) are directly done through SMC. So, it is important to develop the capacity of SMC in environmental monitoring and mitigation. The SMC should ensure that monitoring and mitigation components as spelled out in EMP are properly carried out. The construction of DEO building and other structures when tendered out by DOE/DEO, then the project contractor should be bound by the parameters identified in the environmental and social assessment pertaining to specific mitigation measures in the contract.

c) Bill of Quantities: The tender instruction to bidders should explicitly mention the site specific mitigation measures to be performed, the materials to be used, waste disposal areas, as well other site specific environmental requirements.

d) Allocation of the budget for the building should be as per the design based on geographical setting, remoteness and transportation cost of materials instead of lump sum rates which usually omits the environmental cost.

d) Supervision and Monitoring: The purpose of supervision is to make sure that specific mitigation parameters identified in the environmental assessment and as bound by the contract is satisfactorily implemented. In addition, monitoring is necessary such that the mitigation measures are actually put into practice.

e) Environmental monitoring during project implementation should include systematic assessment of key environmental indicators that determine the degree of environmental impacts of the project and effectiveness of mitigation measures. Such information enables the GON/MOE and the DPs to evaluate the success of mitigation as part of project supervision, and allows specifies the type of monitoring with linkages to the impacts assessed in the EA report and the mitigation described in the EMF.

The monitoring of the construction activities should be done by technical personals (environmental specialist, civil engineers, overseers) from DEO/DoE and or NGOs.

Monitoring activities for project are divided in three types

1. Pre Construction Monitoring
2. Construction Phase Monitoring
3. Operation Phase Monitoring

Pre Construction Monitoring

S.N.	Indicators of Monitoring	Types of Monitoring/ Method of Monitoring	Monitoring Frequency	Responsibility
1.	Printing, Publication & Distribution of EMF to All Stakeholders (DoE, DEO, SMC)	Direct Observation	Once	DoE
2.	Full-time Environmental Officer in place at DoE	Evidence	Once	MoE DoE
3.	Integrating EMF in SIP/ drafting of EMPs	Review of Documents/site observation	Once	DEO
4.	Incorporation EMP in Design, Tender Document	Review of contract documents/Direct Observation	Once	DoE
5	Screening of sub projects/drafting of mitigation measures/or EMPs accordingly	Review of contract documents/Direct Observation	Periodic	DoE

Construction Phase Monitoring

S.N.	Indicators of Monitoring	Types of Monitoring/ Method of Monitoring	Monitoring Frequency	Responsibility
1.	Drinking Water Quality	- Sampling, lab testing & comparison with generic standards - For Arsenic, testing follow protocols mentioned in EFA document	Annual	SMC Note: DEO shall coordinate with district Dept. of water supply & sanitation, NGO, INGOs working in water & sanitation sectors.
2.	Transportation of construction material in covered contains and safe loading & unloading of construction materials.	CM/Direct Observation	Regular during construction	SMC
3.	Water sprinkling in dusty construction area & access earthen roads	CM/Direct Observation	Every Day	SMC
4.	Stockpiling of excavated materials	CM/Direct Observation	Everyday	SMC
5.	Reuse of excavated materials	CM/Direct Observation	Everyday	SMC
6.	Solid waste segregation disposal	CM/Direct Observation	Everyday	SMC
7.	Encroachment of forest	CM/Direct Observation	During construction once in 3 months	DEO/SMC
8.	Quarry & Borrow Pits	Direct Observation	Quarterly	DEO/SMC

Operation Phase Monitoring

S.N.	Indicators of Monitoring	Types of Monitoring/ Method of Monitoring	Monitoring Frequency	Responsibility
1.	Preparation of monitoring reports	Records/Documents	Monthly	DOE
2.	Drinking water quality Arsenic testing and mitigation	Samples taken from different points, source, delivery points Laboratory testing	Quarterly Bi Annual	DEO DOE
3.	Number of orientation and training	Number of orientation and trainings conducted	Regular	DOE/DEO
4.	Impact audit	Compliance with EMF	Annual	DOE/DEO

3.4 Capacity Development

As noted in Section 2.8, environmental capacity is generally very weak at all levels (MOE, DOE, DEO and SMC). MOE, DOE, DEO, and SMC capacity, therefore, needs to be strengthened so that they can take up the responsibilities assigned to them. The engineers assigned for oversight of environmental issues from Physical service section of DOE needs to be trained for monitoring/coordination/compliance of environmental issues related to SSRP projects. GON will implement capacity strengthening measures, such as: i) providing environmental competency/human-resources, ii) training, orientation and awareness activities on environmental planning and management of school and school-facilities, and iii) mechanism for coordination and for accessing specific environmental services e.g. water-quality testing, climate responsive school building construction etc.

If necessary, specific EMF recommends specific, targeted training for the staff, contractors, SMCs to ensure the implementation of environmental recommendations. In addition, the EMF when necessary also provides specific recommendations for operational arrangements to carry out mitigation and monitoring measures. In order to implement the EMF, following is the minimum capacity building necessary.

The capacity development program is designed based on the existing capacity need assessment of the SSRP implementing partners/entities. This has included as one of the sub component under SSRP overall capacity development program.

Capacity Development and Training Program for EMF in SSRP

S. No.	Particular of Activities	Remarks
1.	EMF translation to Nepali, Printing & Publication	
2.	Technical Strengthening of DOE	
	2.1 Recruitment of full-time Environmental Officer	In order to safeguard environmental concerns GON will recruit a separate environmental officer at DOE
3.	- Capacity Building Program to MOE, DOE, (Training, Research & Development) - Monitoring & Supervision Arrangements for DOE, DEO etc)	Detail Capacity Building Program will be Developed by DOE as per need for EMF Implementation

The overall capacity building plan, which forms part of the SSR Plan, will incorporate capacity development for construction, maintenance and rehabilitation related works including the capacity in EMF implementation.

3.5 Integration of EMF into regular programs:

The EMP should be site specific and detailed in its description of individual mitigation and monitoring measures and its assignment of institutional responsibilities. It must integrated into program's overall planning, design, budget, and implementation. The EMP must be prepared if SSRP sub component has to be constructed through tendering.

3.6 Disclosure

The revised EMF will be shared by the MOE with concerned stakeholders After the incorporation of comments and approval from the Bank , this will be translated in Nepali version, printed in several copies and distributed to relevant stakeholders. The revised EMF will be posted in official web site of MOE and Bank's infoshop.

Annexes

Annex 1 - School Selection Criteria

1. 1st Phase: Evaluate all the district level school

The Schools destroyed by wind, fire, landslides, earthquakes & other natural calamities which required new construction.

The school whose classroom is borrowed by resource centre

Community managed schools

Schools running special programs

Based on different educational indicators like enrollment of dalit students (X), enrollment of girl students (Y) ratio of grade 5 to grade 1 students (Z)

$$\text{Index} = 0.4x + 0.3y + 0.3z$$

The school with highest index will get the highest priority & low index as lowest priority.

2. 2nd Phase

$$\text{Adequacy of classroom} = \frac{\text{No of classroom in primary level}}{\text{Total no of classes running in primary level (including sections)}}$$

If, $ACR < 1$, then construction of new classroom & furniture could be provided

$$\text{Classroom Pressure} = \frac{\text{Area of all the classrooms in primary level (m}^2\text{)}}{\text{Total no of students in primary level}}$$

If $CP < 0.75 \text{ m}^2$, then construction of new classroom & furniture could be provided.

$$\text{Student Density} = \frac{\text{Total no of students in primary level}}{\text{Total no of classroom in primary level}}$$

If $SD > 50$, In Kathmandu & Terai new classroom & furniture could be provided

If $SD > 45$, New classroom & furniture could be provided in hilly region

If $SD > 40$, New classroom & furniture could be provided in mountain region

Annex 2 - Chronology of Consultation Meetings

1. Consultation meeting held with DOE/PSS, Kathmandu – Date: 2 April 2009
2. Consultation meeting held at Tanahu District – 17 April 2009
 - a. Shree Satyadevi Lower Secondary School, Bharwa – 5, Tanahu.
 - b. Shree Pabitra Higher Secondary School, Bandipur – 1, Nahala
 - c. Bishnu Secondary School, Nahala, Tanahu
 - d. District Education Office, Tanahu
 - e. Hoste Hainse Child Development Society, Damauli, Tanahu
3. Consultation Meetings Held in Palpa District – 19 April, 2009
 - a. Joint Movement for Children (JMC)
 - b. Adarsha Secondary School, Deurali VDC
 - c. Bhairav Janta Higher Secondary School, Bhairavsthan
 - d. Adarsha Higher Secondary School, Deurali
 - e. Saraswoti Higher Secondary School, Kharsyauli, Palpa
4. Consultation Meeting Held in Pokhara – April 20, 2009
 - a. DEO Office, Bagar, Pokhara
 - b. Sanskrit Secondary School, Bagar
 - c. Bal Mandir Secondary School, Nadipur, Pokhara - 3

Annex 3 - List of Persons Contacted

Local Stakeholders

S.N.	Name/Position	Organization	Remarks
1.	Mr. Damber Bdr. Gurung (Head Master) Mr. Asha B. Lama, SMC, Chairman	Satyawati Devi Lower Secondary School Bhansar, Tanahu	
2.	Mr. Bishwo Raj Adhikari Asst. Head Master	Shree Pawitra Higher Secondary School, Bandipur – 1, Nahala, Tanahu	
3.	Mr. Ram Saran Adhikari	Nahala	
4.	Mr. Min Prasad Nepal, Head Sir	Adarsha Ma. Vi. Laguwa, Deusali	
5.	Mr. Hari Prasad Gyawali	Adarsha Ma. Vi. Laguwa, Deusali	
6.	Mr. Narayan Dhakal	Laguwa, Deurali	
7.	Mr. Rishi Ram Dhakal, Teacher	Palpa	
8.	Mr. Min Prasad Dhakal, Teacher	Palpa	
9.	Mr. Kul Bdr. Kunwar	Palpa	
10.	Mr. Krishna P. Nepal, Teacher	Palpa	
11.	Mr. Bhim Khanal, Educationist	Palpa	
12.	Mr. Tika Khanal	Palpa	
13.	Mr. Lava Pandey	Palpa	
14.	Mr. Binod Kunwar, Educationist	Palpa	
15.	Mr. Dil Bdr. Khanal, Guardian	Palpa	
16.	Mr. Punam G.M., Guardian	Palpa	
17.	Mr. Saroj P. Nepal, Head Sir	Adarsha Higher Secondary School, Deurali	
18.	Mr. Devi P. Adhikari, Guardian	Deurali	
19.	Mr. Moti Raj Adhikari, Teacher		
20.	Mr. Bishnu P. Khanal, SMC, Chairman	Shree Saraswoti H. S. School, Kharsyauli, Palpa	
21.	Mr. Hira Dhar Acharya, Principal	Shree Saraswoti H. S. School, Kharsyauli,	

		Palpa	
22.	Mr. Damodar Bastola, Headmaster	Shree Sanskrit H. S. School, Bagar, Pokhara	
23.	Mr. Cheej Man Gurung, Chairman, SMC	Shree Sanskrit H. S. School, Bagar, Pokhara	
24.	Mr. Bishnu Adhikari, Teacher	Shree Sanskrit H. S. School, Bagar, Pokhara	
25.	Mr. Atma Ram Bastola, Teacher	Shree Sanskrit H. S. School, Bagar, Pokhara	
26.	Mr. Prem Raj Gautam, Educationist	Pokhara	
27.	Mr. Govinda P. Adhikari, Educationist	Pokhara	
28.	Mr. Sushil Basnet, Head Sir	Bal Mandir S. S., Nadipul, Pokhara	

List of District Level Stakeholders Contacted

S.N.	Name	Organization/Address	Position
1.	Mr. Prem Poudel	DEO, Tanahu	DEO
2.	Mr. Hem P. Acharya	DEO, Pokhara	Asst. DEO
3.	Mr. Ghanashyam Adhikari	DEO, Pokhara	Program Officer
4.	Mr. Kulananda Chaudhari	DEO, Pokhara	Overseer
5.	Mr. Dinesh K. Yadav	DEO, Pokhara	Overseer
6.	Mr. Chet Bdr. Thapa	Hoste – Hainse Child Development Society, Tanahu	Program Coordinator
7.			
8.	Mr. Chandra Mishre	Save The Children Norway for DEO School Supervision, Palpa	Engineer
9.	Mr. Rudra Masarang	Save The Children, Norway, Palpa	Program Officer
10.	Mr. Hari Krishna Sharma	Save The Children, Norway, Palpa	Officer
11.	Mr. Ashbir Saru	Save The Children, Norway,	Officer

		Palpa	
12.	Mr. Ganesh Shrestha	Gorkha Foundation, Gorkha	Manager

List of Experts Consulted

S.N.	Name	Organization	Position
1.	Mr. Lava Dev Awasthi	Ministry of Education	Joint Secretary
2.	Mr. Arjun Bhandari	MoE	Joint Secretary
3.	Mr. Mahasram Sharma	DoE	Director General
4.	Mr. Lekh Nath Poudel	MoE	Under Secretary
5.	Mr. Prakash Raj Pande	DOE	Director
6.	Mr. Hari Prasad Lamsal	DOE, Program and Budget	Director
7.	Mr. Shambhu P. Upreti	DoE/PSS	Senior Div. Engineer
8.	Mr. Yuba Raj Poudel	DoE/PSS	S.D. Engineer
9.	Mr. Uddhav Nepal	DoE/PSS	Engineer
10.	Mr. V. Sundararaman	World Bank	
11.	Mr. Mohan Aryal	World Bank	
12.	Mr. Drona Ghimire	World Bank	Environment Specialist
13.	Mr. Shiva L. Bhusal	Embassy of Denmark	
14.	Mr. Ujjwal Man Shakya	Innovative Creators (P) Ltd.	Architect
15.	Mr. Sonam Wangchuk	MS Nepal	

Annex 4 - Summary of Observations from the Field Visit

School Selection Criteria

Most of the schools/SMCs visited during this study period were not satisfied with the school support selection criteria practiced by GON. They felt that the school selection for support should be based on the proper need assessment of individual school rather than generic mechanism developed by DOE. Some of the very old and structurally unsafe school should be include under the priority for support. Some of the schools situated in urban areas with ample spaces could rent their space or building to generate financial resources. Such schools were even found dependent to GON funding rather than being innovative. Such schools should be encouraged to explore the sustainable means on using their available resources to funds for academic purposes.

Financial Support

Most of the schools visited during the study have shown dissatisfaction over the amount of financial support being allocated through GON for the construction of classrooms. The cost of the construction material and labor is very high which doesn't match with cost allocated for classroom construction. The amount of Rs. 10000.00 allocated for improvement in water supply system of school is inadequate.

For 4 roomed RCC building constructed in schools through cost sharing mechanisms (LAGAT approach), the DOE has allocated a budget of Rs. 825000.00. Due to the shortage of areas, schools are attracted towards this approach. Lessons learned from some of the schools with this kind of recipient have clearly shown that the amount allocated is not enough and school needs to plunge in some money on their own to complete the building. This circumstances has created financial crunches in schools.

Construction Materials Used

Though schools are using locally available construction materials (sand, gravel, wood etc) for construction of classrooms, most of the sites visited were found using substandard materials. The sand used was not found of high quality, it was found mixed with soil. The different sizes and shapes of gravel were found used in the construction. In two of the sites visited, unseasoned wood in found used in practice.

The use of wood in hilly areas has been limited to window and doors. However, instead of using tubular iron poles as support, some of the schools were found using wooden post and support. The water is not available in some of the schools visited.

Design and Drawings

The typical design provided by DOE has been followed. There is lack of technical understanding with regards to technical specifications provided by DEOs among the masons, bar benders, and other technicians in schools situation in rural settings. In some schools, innovativeness in construction has been seen. During the construction, spaces for books corner (by providing inner buttress) , visible red stitch band of brick to show earthquake friendly design in construction, well planned landscaping in

limited spaces, smooth edged stone masonry wall construction, installation of rain water harvesting system, use of cement concrete pillar in construction to avoid wood etc.

School safety:

Most of the schools visited were found unsafe as there are no restrictions to enter into the school premises. Due to this toilet doors, water faucet, classroom doors, windows are found broken, wall paintings inside the school are also observed. In some schools, there is no iron grill in windows resulting in burglary and damage of structure by external means.

Though external school improvement is school safety falls under the jurisdiction of school and SMC, very little attention has been paid in this aspect.

School Environment (Orientation, Water supply, sanitation, ventilation, solid waste management)

Orientation: The orientation of school building is satisfactory some exceptions.

Water supply: The amount allocated for water supply improvement is inadequate. Majority of schools don't have water supply systems. Even in schools with water supply system are not properly functioning. Water quality is not satisfactory.

Sanitation: Litters were found scattered in most of the schools visited. Except few schools, sanitation measures were found sub standard. It is due to scarcity of water and habits of students. In addition, the number of toilets proposed in EFA is grossly neglected.

Drainage: Majority of schools don't have proper drainage system.

Solid waste management: 80% of the school visited doesn't have any kind of solid waste management system.

Dust: Due to barren land, lack of vegetation, school grounds are dusty and prone to dust pollution

Noise: The schools visited have CGI sheet roof. This is unsuitable from climatic considerations. The SMC and school teacher, student informed that during summer season, classroom with CGI roof would be extremely hot and making teaching-learning uncomfortable and even torturous for students. During rainy season CGI roof creates noise making teaching-learning impossible. Noise related problem is also observed in two adjacent classes, due to thin non acoustic nature of walls.

Electricity as a backup energy: Majority of schools don't have electric supply.

School sites: Few schools are in landslide prone areas and by the bank of River, and in ridge.

Fencing and external environment: Note satisfactory in majority of schools.

Monitoring Mechanism

The DEO does monitoring of school construction activities in 43 districts. In remaining 32 districts, the DOE has signed an agreement with INGOs/NGOs . During the field visit, it was observed that both the mechanism have pros and cons.

In direct DEO monitoring system, there are no incentives to school inspector so they are not motivated to go for monitoring of school construction activities. In majority of hilly and mountainous region, schools are sparse and the number of personnel, logistics required is found to be inadequate.

The monitoring through NGO/INGO is functioning satisfactorily. It was observed in some schools they are in impression that this kind of monitoring is not related to DEO. There is no regular coordination meeting and interaction among NGO/INGO monitoring team and DEO.

Suggestions and Recommendations:

1. The school shall be designed to be relatively more comfortable in all seasons.
2. The selection of environmentally friendly and cost effective technology should be gradually introduced by DOE.
3. The use of timber in Terai should be restricted and its use in hilly and mountainous areas should be limited to frame of doors and windows. In hills and mountains also this should be gradually phased out.
4. Encouragement of use of local materials has to be tied with quality and monitoring system.
5. The school construction monitoring should be regularize and frequent.
6. The DEO should arrange incentives for their school inspectors in order make the monitoring effective.

Annex 5 - Suggestions from Experts

1. The lessons learned from EFA in implementation of NEGSIFMN should be reviewed (what worked during EFA and What didn't work should be mentioned)
2. The implementation modality, and roles and responsibilities of DOE, DEO, SMC, NGO, and community should be clearly spelled out.
3. The innovativeness and environmentally friendly technology should be gradually introduced in school construction.
4. The environmental requirements in SSRP should be clearly spelled out including the type of environmental screening (if required)
5. The R&D in climate responsive building construction has to be an integral part of SSRP.
6. The linkages with other organizations working in similar field should be established.
7. At district level DEO should work in collaboration with other line agencies like (DWSSO, DOUDBC, etc)
8. The partnership approach with INGO/ NGOs, CBOs should be continued.
9. The capacity building measures for the implementation of EMF in SSRP at different agencies responsible for SSRP implementation should be assessed.
10. The DOE has suggested to tie EMF of SSRP with field reality and in consistent with the improvements in the existing practices and it has to be implemented based on "ground reality". Non achievable and very ambitious thing should not be included in the EMF.

***Annex 6 - Government's Effort towards School Physical Facilities
Development through various programs (update up to 2007/2008)***

S. No.	Programs	New Classroom	Rehabilitation of Classrooms	Toilet & Water Supply	Maintenance Training
1	SERDP (1981)	941	1,170		
2	PEP (1985-1990)	951	1,047		
3	EAARRP (1988-1996)	15,578	0	0	0
4	PEDP (1990)	2,392	0	0	0
5	BPEP – I (1993-1998)	11,154	7,122	191	4,854
6	BPEP – II (1999-2004)	5,392	10,794	5,093	4,860
7	EFA (2004 – 2009)	18,000	5,810	3,495	0
8	JICA (1994 – 2006)	8,028	0	1,747	0
9	Others	4,284	2,217	0	0
	Total	66,720	28,160	10,526	9,714

Annex 7 – DOE Policy

A. DOE Policy for Improvement of School Physical Facilities (Classrooms, Toilet & WS)

- **Decentralization (Construction through Community)**
 - Need Identification through community
 - Program implementation through Community
 - Ensuring ownership & thereby ensuring sustainability of the created facilities

- **Collaboration with I/INGOs – (Out of 75 Districts – 32 Districts – SCNN-20, Plan International-6, Room to Read – 3, SCUS-2, SCJ-1)**
 - Harmonization
 - Quality Supervision
 - Involvement of Social Mobilization
 - Better Transparency (Public Audit)
 - Effective in awareness Raising

B. DOE Policy on Design & Construction of School Building

- **Structural Safety of buildings**
 - Design and construction as per National Building Code
 - Safe against Earthquake and strong storms
 - Construction supervision by qualified Engineers/Sub-Engineers.

- **Child-friendly design & construction**
 - Classrooms (0.75 sqm per children, T:50, H:45, M:35 students per classroom)
 - Furniture (Floor seating for grade 1 and 2)
 - Water supply facilities
 - Separate toilets for girls & boys

- **Environment-friendly design & construction**
 - Enforcement of *National Environmental Guidelines for School Improvement and Facility Management in Nepal*

- **Inclusive design & construction for disables**
 - Ramp construction for wheel chair movement
 - One bigger door in every classroom & toilet for wheel chair movement

- **Community contribution in facility development**
 - Creates ownership feeling
 - Ensure sustainability of the created facilities
 - Ensure regular repair & maintenance

Annex 8 - Environmental Screening

Checklist

Medium/Area	Screening Question	Yes	No	Possible (-ve) Impact
Slope Stability	- Are there any landslides or other forms of mass instability on the slopes? - Is there erosion or gully development?			
Spoil Disposal	- Can spoil be reduced? - Is spoil being tipped away from designated areas? - Is spoil falling or being washed on to farmland?			
Water Management	- Are slope drainage designs adequate? - Are drainage outfalls unprotected against scour and erosions? - Is there any disruption of drinking or irrigation water supplies?			
Land Use	Has there been any loss of land for which landowner should be compensated?			
Plants and wildlife	- Are large numbers of trees being removed? - Is any form of wildlife being disturbed?			
Stone crushing and?	Is the project operating stone crushing?			
Hazardous Materials	- Is there project using any type of hazardous material (e.g. bitumen, cement paint, explosive fuels, lubricants etc) - Laboratory chemicals			
Dust	Is dust generated from construction works or by construction traffic?			
Renovation	Will construction, renovation work disturb other academic activities?			
Noise	Are there increased traffic noise from construction and transportation of vehicles and machinery?			
Water supply and sanitation	Will it temporarily stop water supply and sanitation system?			
Ecosystem	Will the renovation work have any indirect impact on environment and ecosystem?			
Surface water	Does construction of pit latrines, septic tanks and soak ways could cause seepage of contaminated water into aquifers			
Ground water	Is there any pollution of Ground water from improper practice by SSRP school			

	construction activities?			
Laboratory safety and hazardous waste	• Are there pollution from laboratory waste disposal • Injuries from poor chemical handling			Applicable to schools with recipient of laboratory support under SSRP
Health and Safety	• Are there any issues related to health and safety issues?			
Social Issues	• Does construction involve local people, skills and material? • Does contractor have grievance addressing mechanism? • Does project built up rapport with local people and community? • Are there conflicts between project and the local people?			
Use of construction materials	• Does construction material (prescribed) have lowest environmental impact?			
Material specification	• Use of locally available suitable material, practices of reduce reuse, recycle of materials during construction.			

Annex 9 - The Design Guideline for School Buildings Construction from Environmental Perspective

6.1 Background

The technical-fix for achieving a healthy environmental solution depends on different variables. This chapter proposes the practically achievable options through the understanding of different combinations of those variables.

A host of interrelated considerations—from site design to transportation issues—factor into the creation of a "high-performance" environmentally friendly school. To effectively integrate environmentally friendly strategies, these options must be evaluated together from a whole school building perspective early in the design process. To do this, school will required to form a team of players from the community and school to work in collaboration with design team. The school design team must have expertise in integrating energy and environmental considerations into school design. Inputs from all stakeholders factor into good functional design to ensure the environmentally friendly construction of school.

The following list provides basic information about some key elements to consider when designing a high performance environmentally friendly school building. District education office and Department of Education include these elements in their Request for Qualifications (RFQ) or Request for Proposals (RFP) for design professionals. Once a design team is selected, comprehensive design guidelines available through Department of Education can provide the team with detailed information on the following strategies.

6.2 Commitment and Motivation

There is no doubt that the single most important factor in achieving environmentally friendly school is the presence of committed and informed people. The emphasis should be given on the commitment. People are often well aware of the health risks and the theories of contamination but do not act on that knowledge.

There is no simple formula for making people committed. However, recognizing and valuing people's effort, and ensuring that there is sufficient scope for their own decision-making and creativity can go a long way towards encouraging sustained commitment. This point is particularly important since, it argues against standard designs.

The success of efforts in promoting environmentally friendly school relies on the awareness and commitment of the community to want such a program and to ensure that it is maintained within the community. The collective recognition and understanding of the problem by families and community members can help to bring about support and action for ongoing efforts to influence behaviors and conditions that will improve the environmental situation in school

Community commitment is strengthened/marked by

- Acknowledgement of the importance of the problem from education, environment and other groups
- Allocating local resources, e.g. public money for incorporating environmentally friendly products in schools
- Combining the program with other programs in the community, such as primary health care and sanitary environment
- Ongoing efforts to attract media attention
- Involving existing councils, school committee, organizations such as women's group, NGO, Eco clubs etc
- Use of both problem solving as well as environment promoting approaches to improve school environment

6.2.1 Supportive school policies/practices: Development of supportive school policies and practices related to environmental improvement is improvement to make a program sustainable. For example; regulations for improving and maintaining adequate sanitary facilities and a safe water supply at school are essential. Also, collaboration and coordination between the education and environment sectors, and between the school and the community are important requirements for success.

Examples of supportive policies and regulations

- Required periodic coordination between environment and education authorities at local, district, and national levels
- Rules about cleanliness for students and personnel, and about the proper use of latrines and water supply facilities
- Rules about keeping the school environment clean

Examples of supportive practices in schools

- School curriculum includes environment/environment preservation
- Proper use of hygienic toilets/latrines
- Regular cleaning of toilets/latrines
- Existence and use of proper hand washing facilities
- Safe food and water
- Formation of eco clubs, environmental awareness campaigns in schools
- Teacher's training on environment related subjects
- School/community projects and outreach

6.3 School Building Siting

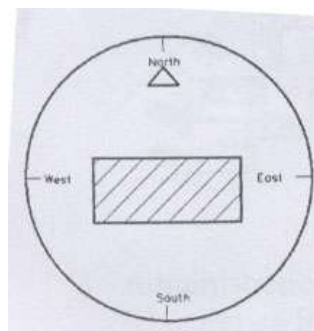
6.3.1 Orientation: School building orientation is very important as it determines magnitude, natural heating, and lighting available inside the building. Orientations are based on the climate type of the

place where school facilities are planned. In Nepal three geo-zones are available viz: Terai, Mountain, and Hill. The orientation of school building in each of the above type is described below.

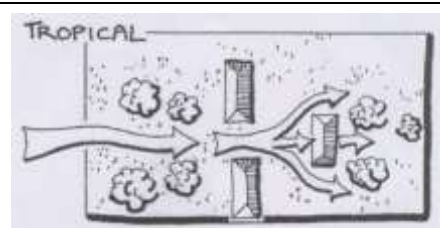
Terai Region

Arrange the school buildings in such a way that the prevailing winds can be used in cooling rooms.

The school buildings should be oriented facing between West to North. In case, the building can't be avoided facing East, West or South, evergreen trees have to be planted to prevent direct sunlight entering the building.



Orientation criteria sometimes are influenced by the change in the school timing. For instance, during the summer in some parts of the Terai, there is a practice of conducting the classes early in the morning (6.30 hrs to 10.30 hrs with no break); in such a particular case North or North-East facing causes direct glare in the class rooms; other side of the coin is that in such a case and when the site does not permit the ideal orientation, orientating the classrooms towards West-West-South supported by other measures may be permitted.

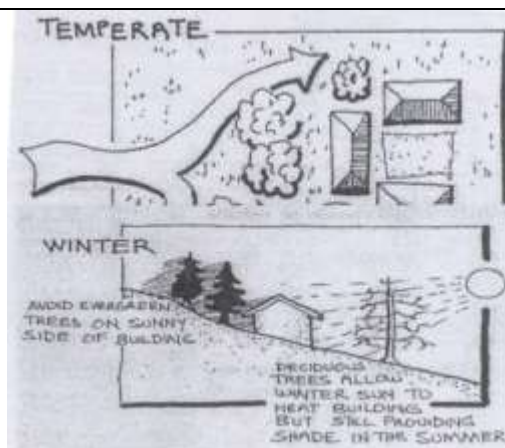


Mountain Region

In the mountain region, making use of sun to heat the rooms is a good design concept. Therefore, building orientation should be between East and West. One-way to blocking the seeping winter wind is to plant trees in the direction of the prevailing wind.

Hill Area

It is advisable to provide deciduous trees and at the same time allow winter sun to heat up the buildings.



Understanding the location and the surroundings including the contour is very important for developing the Master Plan. A building that looks good and fits well at one site may not necessarily

look good at another site. Poor site selection for school buildings can lead to high costs on account of earth retaining structures. A careful study of the terrain can easily help to reduce this as shown in the illustration and table. For any given location, a site that results in the minimum cut height “h” should be selected.

6.2.2 Site Selection and Planning: Both, site selection and site planning have a major input on the relative "greening" of any school facility being planned. Site selection for school includes such issues as transportation and travel distances for students, impacts on wildlife corridors, and impact on the hydrology. Decisions that school make during the site planning will impact the immediate natural community as well as the school building, energy consumption and student comfort. Good site planning minimizes site clearing; preservation of existing vegetation may provide a low maintenance landscape and avoids supplemental irrigation. Mature stands of native vegetation in school compound often provide the desired energy conserving shade and wind control that would otherwise required years to develop for expensive new plantings. Thoughtful placement of a school building on a site promotes energy conservation by taking advantage of natural site features such as topography, sunlight, shade and breezes.

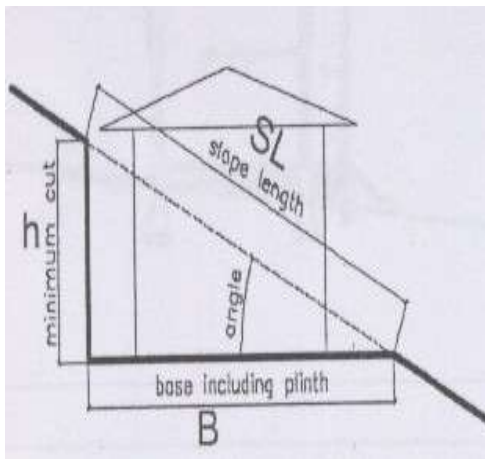
While selecting the site or laying the school building, it is very important to note of features like religious objects, protected forests, river, stream, landslides etc. within and around the site. These are additional factors that need attention apart from other factors.

It goes without saying that school building should protect its occupants from rain, wind, sun and snow. But it need only offer, as much protection is necessary. A classroom of a standard requirement in mountain regions may be quite unnecessary and inappropriate in a tarsi region. Schools should be designed to prevent extremes of temperatures inside classrooms.

6.2.3 Approximation for earth cutting and retaining wall The Site Development Requirements form the Perspective of Earth Cutting, Filling, and Retaining Wall Structures:

Schools in mountain and hill region of Nepal fall on slopes. Poor site selection for school buildings can lead high cost on account of earth retaining structures. A careful study of terrain can easily help to reduce this as illustrated in the table. For any given location, a site that results in the minimum cut height “h” should be selected.

	Small Multipurpose hall			Large Multipurpose hall		
Angle	B	H	SL	B	H	SL
5	19.7	1.72	19.78	24.3	2.13	24.39
10	19.7	3.47	19.97	24.3	4.28	24.63
15	19.7	5.26	20.32	24.3	6.49	25.07
20	19.7	7.17	20.97	24.3	8.85	25.86



Large multipurpose hall

25	19.7	9.18	21.72	24.3	11.32	26.79
30	19.7	11.37	22.73	24.3	14.02	28.04

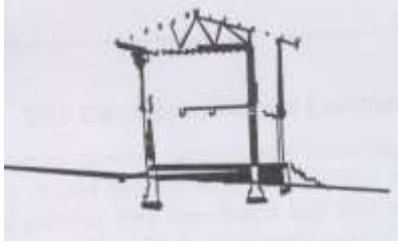
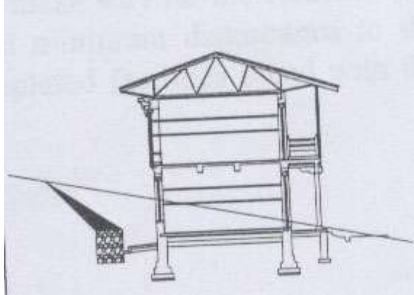
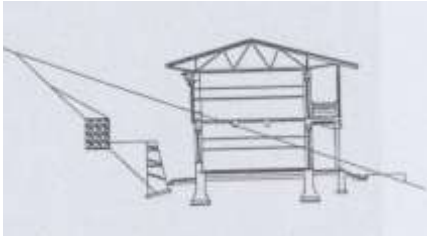
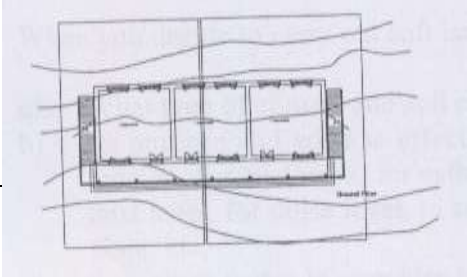
B	H	SL
24.3	2.13	24.39
24.3	4.28	24.63
24.3	6.49	25.07
24.3	8.85	25.86
24.3	11.32	26.79
24.3	14.02	28.04

	Academic buildings		
Angle	B	H	SL
5	11.7	1.02	11.74
10	11.7	2.06	11.86
15	11.7	3.12	12.07
20	11.7	4.26	12.45
25	11.7	5.45	12.90
30	11.7	6.75	13.50

This table will help to guide during the site selection and site development.

The following illustrations show some of the real site situations. The Master Plan may not show the full details of earth cut and fill and wall construction. The site engineers can refer to these illustrations to find optimal solutions.

Site situations and earth cutting

Here the ground is at 5° slopes and it is cut to fit the building. As such, no retaining wall is required if the cut surface is riveted with vegetation.	
In this case, the building is placed with minimum disturbance to the ground. This is possible if the slope of the ground is less than 5°. However, some filling may be required to bring the rooms in one level.	
Between 10° and 15° slopes, massive retaining wall is not necessary. Study the topography of the site and go for low height retaining or gabion wall. If the ground is not loose or marshy, the heights of these walls can be of half the cut height. The upper half height can be manipulated as per the table given depending on the angle of repose. If a gabion wall is built, cover the top of the walls with good soil and plant greenery for beautification.	
For a grade between 15°-20°, the height of the retaining wall can be approximately $\frac{2}{3}$rd of the cut. However, as the ground becomes steeper to 20°, this rule-of-thumb may not be applicable. In this case, follow the illustration given below.	
In such a situation, a split retention wall is used. When this type of wall is built, it should be ensured that the active resultant force of the upper retaining wall falls on middle third of the lower wall foundation.	
Here, the left half of the building falls on a steep slope as compared to right hand side half. In such a situation, one option can be the use of combined single and split retention wall or adopt the cut and fill	

method on the left portion, and only cut method on the right. However, before making the decisions, economic advantages have to be considered.	
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In this case, the building is placed with minimum disturbance to the ground. This is possible if the slope of the ground is less than 5° . However, some filling may be required to bring the rooms in one level.

Between 10° and 15° slopes, massive retaining wall is not necessary. Study the topography of the site and go for low height retaining or gabion wall. If the ground is not loose or marshy, the heights of these walls can be of half the cut height. The upper half height can be manipulated as per the table given depending on the angle of repose. If a gabion wall is built, cover the top of the walls with good soil and plant greenery for beautification.

For a grade between 15° - 20° , the height of the retaining wall can be approximately $\frac{2}{3}$ rd of the cut. However, as the ground becomes steeper to 20° , this rule-of-thumb may not be applicable. In this case, follow the illustration given below.

In such a situation, a split retention wall is used. When this type of wall is built, it should be ensured that the active resultant force of the upper retaining wall falls on middle third of the lower wall foundation.

Here, the left half of the building falls on a steep slope as compared to right hand side half. In such a situation, one option can be the use of combined single and split retention wall or adopt the cut and fill method on the left portion, and only cut method on the right. However, before making the decisions, economic advantages have to be considered.

Both site selection and site planning have a major input on the relative "greening" of any school facility being planned. Site selection for school includes such issues as transportation and travel distances for students, impacts on wildlife corridors, and impact on the hydrology. Decisions that school make during the site planning will impact the immediate natural community as well as the school building, energy consumption and student comfort. Good site planning minimizes site clearing; preservation of existing vegetation may provide a low maintenance landscape and avoids supplemental irrigation. Mature stands of native vegetation in school compound often provide the desired energy conserving shade and wind control that would otherwise required years to develop for expensive new plantings. Thoughtful placement of a school building on a site promotes energy conservation by taking advantage of natural site features such as topography, sunlight, shade and breezes.

While selecting the site or laying the school building, it is very important to note of features like religious objects, protected forests, river, stream, landslides etc. within and around the site. These are additional factors that need attention apart from other factors.

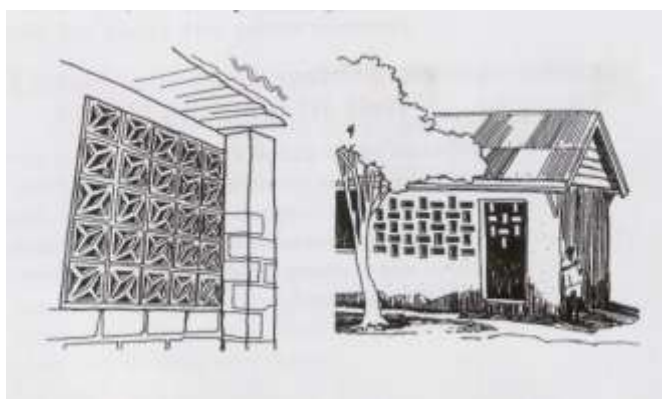
6.3 Day Lighting and Windows

Poor indoor lighting in classroom can have many harmful effects on health and well being of teachers and students. A poorly lit classroom can lead to eyesight problems. Poor lighting within the classroom can also make students feel more depressed. Generally, these problems can be remediate by adding windows to the classroom to increase the amount of natural light, however because of different climatic condition of Nepal, window selection should be based where facility is planned. Increasing natural light is also important for classroom cleanliness; if a classroom is dark, it is more difficult to see dust and dirt and thus more difficult to clean properly.

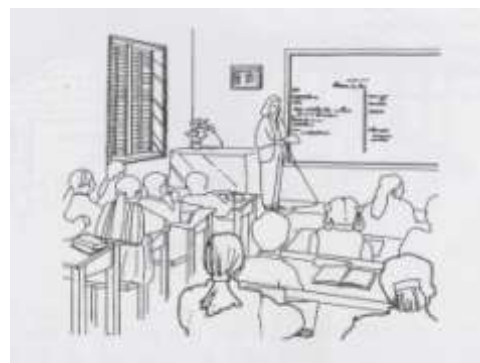
Increased use of daylight helps to reduce electric lighting usage, and avoiding the heat generated by lighting fixtures. High performance windows with low e-glazing also help to minimize heat gain in warmer months and heat loss in colder.

It was noted that in Terai of Nepal, people try to increase ventilation, which may result in a teacher standing against the glare from an open window. In mountain, because of cold, the tendency is to build small window resulting in dark classrooms.

An alternative to large windows is the perforated screen wall. This is a solid masonry wall punctured by numerous closely spaced holes through which light can filter. If the wall is built of bricks the perforation effect can be achieved by simply leaving spaces between the bricks at regular intervals. If it is made of concrete (or earth-cement) blocks, special moulds can be used to produce blocks with decorative holes. Where masonry is not used for construction, or where there do exist large window openings, other kinds of screens can be created, for example using horizontal bamboos.



A perforated window is an alternative to a window opening in warm climates. It provides security and relatively even distribution of light.



A louvered window could provide glare-free light at the front of classroom

The advantages of perforated screen wall are that it provides security and a relatively even distribution of light. The disadvantages are that it does not protect against wind, cold and dust, and classroom (particularly if large) can remain rather dark unless the internal walls are also light colored. Care must be taken to orient the building so that direct sunlight does not penetrate the screen wall, since this will almost certainly lead to glare.

In mountains, where small windows are necessary to reduce heat loss, it makes sense to maximize the light which light that can enter through a small opening. A roof light lets in considerably more light than a window of equal size in a wall. A provision of sky light in CGI roof is found in schools in mountain. This arrangement helps to bring more light into the center of the classroom, which is often poorly served by wall windows. However, care has to be taken to avoid direct sunlight falling onto desks.

It is also important to make the best use of any light once it has entered the room. Untreated mud-and dung plaster on the walls, for instance will make a room relatively dark. The underside of a thatched roof will absorb lot of light. The level of light can be dramatically increased with a light colored ceiling and lime wash on the walls. These can also reduce glare by ensuring that light is reflected from all directions. In addition, adoption of matt-finished blackboard and regular repainting it can be relatively cheap way of mitigating the impact of poor lighting.

6.4 Renewable Energy Systems

Use of solar and wind technologies with battery storage can provide emergency power supply, or a source of steady power. Renewable energy system in school can increase maintenance requirements, but they also reduce utility costs.

Some schools located in hills of Nepal have no electricity. So for the lighting and other purposes like operation of basic science equipment, computer, it is very necessary to find alternative means of electricity. So renewable energy could become an important element for meeting school energy demand.

In mountains of Nepal, majority of children don't take shower because of unavailability of hot water. Even school with necessary infrastructure and other facilities lacks this facility. Introducing solar technologies for water heating purposes could solve this problem.

6.5 Ventilation Systems

Adequate school ventilation is important. The poor ventilation in schools can lead to respiratory problems, and make tuberculosis and easier transmission of infectious diseases. Constructing schools with a sufficient number of windows may improve ventilation. Alternatively, schools can be constructed using bricks with holes drilled through them ("air-bricks"), which allow fresh air to circulate within the classroom.

Using the whole-building approach, school designers can factor in energy saving choices that reduce heating and cooling loads.

6.6 School Sanitation

The high incidence of disease, particularly among children is largely attributed to unsafe drinking water, poor personal hygiene practices and unsanitary environment. Each and every school should have adequate safe water supply system and sanitation facility. But majority of rural schools in Nepal neither have proper water supply nor sanitation system, particularly toilets.

6.6.1 General Criteria for Hygienic Facilities and Safe Water Supply in School: There are many methods that can be used to provide hygienic sanitation facilities and a safe water supply at school.

The principles and criteria involved in the selection of appropriate methods stipulate that those chosen should be:

- Technically and environmentally sound
- Financially affordable
- Socially and culturally acceptable
- Reliant on labor and resources available in the community
- Simple to install, operate and maintain
- Easily accessible by the students
- Related to reducing public health problems that are perceived as priorities within the community

6.7 Drinking Water

A "safe water supply" is a source of water that is not contaminated by dirt, bacteria, parasites, or anything else that could cause contamination.

To promote school health an easily accessible water supply should be available that provides sufficient safe water to meet school needs. School water needs, can be estimated by questioning stakeholders of school about their daily water use. If this is not possible, minimum water need should be calculated. To ensure that the water is potable, either the water supply should be protected or the water should be treated before use. Often, unprotected water sources such as springs, traditional wells and ponds, can be improved and this may be preferable to constructing new supplies. However, unprotected sources are open to contamination and pose a potential health risk. School environmental health program should therefore promote the use of protected drinking water source.

Some countries have set minimum standards for the supply of safe drinking water to pupils. For instance, in Viet Nam, every school is supposed to provide 0.33 liters of boiled water per pupil per day in summer and 0.1 liters in winter.

Characteristics of low-risk water sources

- The water source is fully enclosed or protected (capped) and no surface water can run directly into it.
- People do not step into the water while collecting it.
- Latrines are located as far away as possible (minimum 15 meters away) from the water source and preferably not on higher ground. If there are community concerns about this, expert advice should be sought
- Solid waste pits, animal excreta and other pollution sources are located as far as possible from the water source.
- There is no stagnant water within 5 meters of the water source
- If wells are used, the collection buckets are kept clean and off the ground or a hand pump is used.

The water source protection technique is mentioned in annex 2.

Checklist for schools considering water supply improvements

- *Have stakeholders of school been fully consulted about the type of water supply?*
- *Have stakeholders of school had previous experiences with water supply improvements and have these been relayed to the relevant agency?*
- *How will the water supply be managed to ensure that it is reasonably accessible to everyone in the school?*
- *What are the long-term financial implications of the choice of water supply?*
- *Can the school afford to pay expected operation and maintenance cost?*
- *What spare parts are required and how often should they be replaced?*
- *Who will be trained to operate and maintain the water supply?*
- *Will the quality of the water be tested?*

Among the many types of water sources schools in Nepal are found using the sources like protected springs, dug wells, boreholes, and piped water supply.

6.7.1 The Available Types of Water Sources in Nepalese Schools

6.7.1.1 Protected Springs: A spring is where underground water flows to the surface. Springs may occur when the water table meets the ground surface, these are called gravity springs. In some cases, ground water is held under pressure and spring come to the surface because of a natural break in the rock, or because a shallow excavation is made. Springs can make very good water supplies provided they are properly protected against contamination. If springs are found above the school, they can feed a pipe system for providing water close to school. When a spring is at the same or lower level than the school, it can still be protected, but greater care is needed and it is unlikely that water will flow through the pipe system by gravity. The first step in deciding whether a spring should be protected is to determine whether it provides enough water for expected number of students. Measuring the time it takes for the spring to fill a bucket of known volume easily does this.

6.7.1.2 Dug wells: Dug wells are usually shallow wells dug by hand, although some may be quite deep, and they are often lined with bricks. However, unless artesian water is tapped, many dug wells go dry or have very little water in dry periods because it is difficult to sink wells below the water table without using more sophisticated techniques.

6.7.1.3 Boreholes: Boreholes are narrow holes drilled into the ground that tap into groundwater. Generally as the borehole is drilled, a lining of plastic, steel or iron is sunk to protect the hole from collapse. The lining has slots in the bottom of section to allow entering the borehole and gravel is placed around the bottom of lining to improve flow and provide filtration. The top few meters around the borehole should be sealed using concrete, and a concrete apron is cast around the top of the borehole to prevent surface water from flowing into the lined shaft. A stand is usually cast into the apron to provide a stable base for the pump. Once the borehole is completed it should be cleaned with chlorine and the pump installed. Majority of schools in hill and tarai of Nepal are using boreholes with handpumps. Boreholes usually provide good quality water, but the water sometimes contains harmful chemicals, such as fluoride and arsenic, or nuisance chemicals such as iron, etc. The school should

carry out chemical analysis of water in a regular interval as discussed under the monitoring protocol of this guideline.

6.7.1.4 Piped water supply: Many schools in Nepal have piped water systems. These piped water systems are often small and rely on community management. Most piped water supplies include storage tanks so that water is always available, even when demand is heaviest. Such tanks are usually necessary because the rate of water use at peak times of the day in school is greater than the average rate of use throughout the day. The tanks also provide emergency storage in the event of a breakdown. Piped system in a school requires regular maintenance. Pipe leaks likely to be used heavily by student. As a result, the taps are more likely to break and will need frequent replacement. One-way of dealing with these issues is to give someone in the school responsibility for school taps and making repairs. To prevent the accumulation of stagnant water around school taps, school could build a concrete “apron” at the base of taps and include a drain and soakage pit.

Another problem with piped systems is that students do not consider the impact of how much water they use, and may not think it is important to turn off the tap after use. When there is a lot of water, this may not have negative consequences. However, where the amount of water available is limited, there will be shortage of water. Schools with piped water systems should thus be aware of these impacts of their water use and good water use should be promoted.

need to be repaired rapidly to prevent water loss, and to prevent surface water from entering the pipes and contaminating the supply. Also taps are

6.8 Rainwater harvesting scheme in schools

Although rainwater can be a good source of water for drinking and other purposes in school, it may be seasonal, and it is often difficult for a school to rely on rainwater alone. This technology could be used in conjunction with other available sources of water. Schools for their own use could collect rainwater. If the rainwater is to be used for drinking purpose it is better to collect it from a roof, rather than from a ground catchments where it may become contaminated. Ground catchments are more appropriate for use in garden and watering of grounds.

Using roofs to collect rainwater is relatively easy and a lot of water can be collected. For example; 50 mm of rainfall on a 4-m² roof yields 200 liters of water. All that is required are gutters around the roof that discharge into a collection tank. The roofing material is important and hard surfaces, such as iron sheets or tiles. Hard surfaces are also easier to keep clean and are less likely to have insects and animals living in them.

Any roof used to collect rainwater for human consumption must be thoroughly cleaned at the start of the rainy period. Birds and animals may leave faeces on the roof and these can be a source of pathogens. There should be a system for diverting the flow of water in gutters away from the tank, so that the first rain (which are more likely to pick up contamination from the roof) are not collected. A small filter may be added to the top of the collection tank as an added protection. The tank should also be cleaned every year and any silt or algal matter removed. After cleaning and before use, the tank should be scrubbed using a chlorine solution (bleach).

Water should be drawn from a tap at the base of the tank, rather than with a bucket, which may contaminate the water. It is better not to bury the collection tank, even partially, since contaminated

water from the soil can enter the tank. Covering the tank is also essential for preventing contamination of the water and for reducing opportunities for disease vectors to breed.

Protection of above mentioned water sources that could be used in a school in Nepal is a matter of discussion in this guideline. The protection measures for different water sources are highlighted in

6.9 School Water Treatment and Safe Handling of Water

The best option for improving the school water quality is to treat water in the school by filtering, chlorinating, boiling or leaving the water to settle. Individual school could select the options as appropriate. These options are discussed in more detail in following sections.

6.9.1 Boiling: Bringing water to a rolling boil (20 minutes) will destroy pathogens in the water and make it safe to drink.

6.9.2 Candle filters: Candle filters are hollow, porous ceramic cartridges, commonly found used in schools of Nepal. Although they do not filter out all pathogens, they remove the larger ones such as protozoa, worms, and bacteria (but not viruses). Ceramic candles need careful maintenance and should be cleaned and boiled at least once a week, even if they are not clogged. If a candle filter becomes clogged, it should be scrubbed under running water with a stiff brush free of soap. To reduce the risk that water will pass through a candle without being filtered, such as through small crack, candle filters should be regularly inspected and replaced if necessary.

6.9.3 Chlorination: This will kill most bacteria and some viruses. Since that taste of chlorine disappears when water is left in open containers, very small lump of bleaching powder can be added to a 20-liter water container and the mix left to stand for at least 30 minutes. After this time, if a faint smell of chlorine can be detected in the water, it should be low –risk and palatable to drink. Chlorine should only be added to clear water otherwise the dirt in the water would absorb it. Other disinfections systems that have been developed for treating water, particularly through the use of solar radiation. This technology is commonly known as “SODIS” in Nepal. This is effective in treating water, although this may take longer than chlorine disinfections.

Frequently water collected from a communal point and transported back to school for use becomes contaminated because of poor handling. Stakeholders of school should therefore be aware of the risks of contaminating the water and how it can be prevented.

6.10 Water Conservagtion:

Although it is important that students use enough water for good hygiene, in areas where water is scarce it is also important not to waste water. As mentioned above, piped water supplies are particularly vulnerable to wastage, if they are not properly managed, the surrounding community of school may suffer water shortages. Most piped water systems leak and need to be checked regularly and repaired as soon as faults are discovered. Taps should be turned off immediately after use and students discouraged from playing with taps.

Key Points

- Functional water pints providing adequate supplies of safe water are rarely found in schools
- Hand washing facilities are very rarely observed in schools

- Although environment and hygiene education forms part of curricula in Nepal, teachers have usually not received adequate training in those areas
 - Other environmental issues weigh heavily on the school community, disposal of solid waste and repairs to school buildings, inadequate infrastructures to support basic environmental practices.
- The methods of protecting different water sources used in schools are shown in annex 2

6.11 Sanitation Facilities:

Safe disposal of excreta, so that it doesn't contaminate the environment, water, food, or hands is essential for ensuring a healthy school environment and for protecting personal health. This can be accomplished in many ways, some requiring water, others requiring little or none. Regardless of method, the safe disposal of human faeces in school is one of the principal ways of breaking the faecal-oral disease transmission cycle. Sanitation is therefore a critical barrier to disease transmission.

One reason why latrines are often out of order in Nepalese schools is that they are frequently inappropriately designed for use by children to use. Latrines in Nepalese schools vary in design from communal three to six hole latrines served in schools in Sindhupalchowk, to simple pit latrines. The number of latrines available/observed in schools during the field visit was without exception insufficient to meet the needs of students, particularly at time of peak usage such as break. The situation contrasts with the standards generally accepted within the health, environment and education sectors, which are shown in box discussed below.

The provision of *one latrine for approximately thirty pupils* is generally recommended, if urinals are also available. This represents *one latrine for every twenty girls* and *one for every forty boys*.

When planning the number of latrines for a school, certain issues should be considered:

- Are separate urinals available for boys? If so, fewer latrines will be needed.
- What is the proportion of boys to girls? If urinals are available, boys need fewer latrines.
- Are children allowed to leave the classroom during classes to use the latrine? If not, the pressure on latrines at break/recreational time will be great and more latrines will be required.
- Do all children have breaks from classes at the same time? If so more latrines will be required. Could breaks be staggered?

6.12 Drainage:

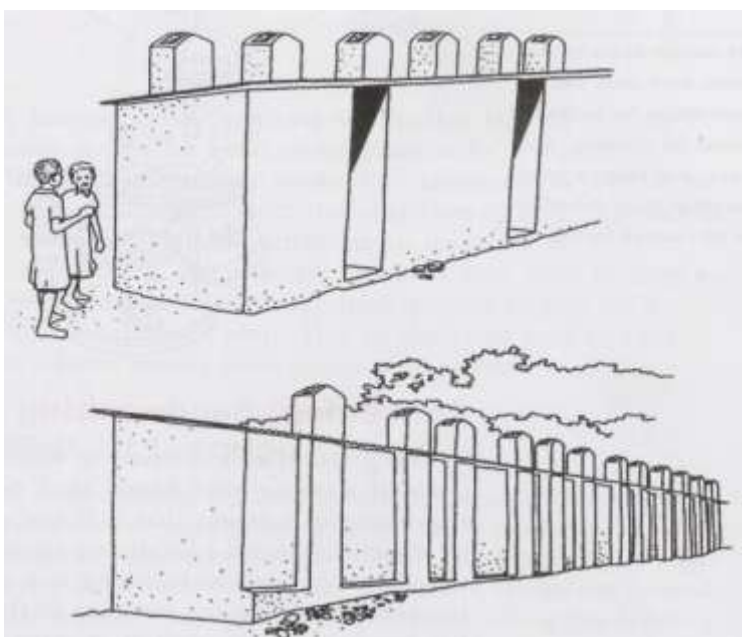
The planning and implementation of proper drainage system is very important. The following guidelines must be considered for drainage designing and in its construction.

- The design should be as simple as possible considering for future development
- Generally, if the drains are not wider than 150X 150 mm, it can be left open but any drain bigger than 150 mm should be covered.
- Drains carrying off surface water can be open or covered
- In both the cases, silt traps at appropriate distances should be provided. However, if there is sudden level difference created by walls or land profiles, two silt traps may be required
- General rule of thumb is to provide silt traps at every 20 meters intervals for easy cleaning.
- All drainage should be self-cleaning and should function at minimum running maintenance costs.
- The main collection drain should always be covered or laid under ground.
- Normal roof drains are 150X100 mm at 5% slope. The slope and depth will increase as the roof drain reaches the outlet of the collecting drain.

- Minimum size recommended for the collecting drain is 300 X250 mm and should be covered with concrete slab with perforation to take in surface runoff waters
- The depth of the collecting drain will depend on the local and distance from the sub-drains, such as roof drain, retaining wall drains, etc.
- While collecting and discharging surface waters, it should not cause public nuisance or pose danger to health and property.
- Beside the roof drains, it is an advantage and desirous to construct surface water drains around the school buildings, playgrounds, open spaces etc to prevent flooding during the monsoon.
- It is not advisable to seal off the manhole and inspection chamber with concrete completely if intermediate ventilation is not provided. This will cause cracks of pipes, disintegration of joints etc
- Any drain provided around or within the courtyard must be concealed.

6.13 Technologies for excreta disposal in school

The off set latrine system is economic, easy in design, could be constructed with local available materials and easy to built and maintain for schools. Though the choice of technology to be adopted



for toilet purely depends on the water availability, budget, soil type etc. This particular type is found very useful to recommend as standard design in Nepal.

This latrine do not require periodic emptying, once a pit is full it is sealed and a new pit is dug. The faecal matter is left to decompose in dry conditions for at least two years, the contents can be safely and the pit reused. These latrines could also be designed to allow faecal matter to compost and be reused in agriculture. Other designs that could be incorporated are two alternating

pits, reducing the need for new pits. As shown in design, ventilation is installed to remove odor and flies, basic and locally available materials could be used. Mostly the toilets constructed in Nepal were problem of chocking; students throw carelessly all the solid waste (paper, plastics etc) in toilet resulting in blocking the trap. As per the standard, the number of toilet required for each school will be calculated, and the toilet is proposed in series as shown above. The urinals could be built to reduce the number of toilets required emptied manually

One main reason for poor maintenance of the school toilets was due to separate toilets for staff and student usages. These arrangements create less emphasis and inadequate monitoring of student's toilet. This was observed even in one of those schools where there was plenty of water available for maintaining the toilet. It is suggested that in order to improve the sanitary condition of toilet, and to gain students confidence towards cleanliness, all the stakeholders of school should share toilets.

For a school latrine to function properly it must be maintained and cleaned on a daily basis. Somebody must have specific responsibility for this and compliance must be checked. Groups of school children might do the cleaning in rotation.

Annex 10 - Laboratory Checklist from Environmental Perspective

S. No.	Screening Question		Yes	No	Remarks
1.	Will the research academic work be laboratory based? [If the answer of question 1 is No, then go to question 4]				
2.	Does the laboratory have				
	i	environment, health and safety protocol or guidelines?			
	ii	adequate fire safety provision?			
	iii	safety provision for gas cylinder handling?			
	iv	proper waste disposal facilities?			
	v	adequate liquid waste management facilities?			
	vi	proper storage facilities for hazardous chemicals, pesticides etc?			
	vii	adequate ventilation system?			
	viii	first-aid facilities?			
	ix	emergency exit facilities?			
	x	trained professional to guide the researchers/students about safety procedures?			
3.	Will the laboratory based research work				
	i	produce hazardous waste materials?			
	ii	generate infectious waste?			
	iii	cause significant emissions of gas harmful to ions of gas harmful to health?			
	iv	generate liquid waste?			
	v	cause any major noise?			

S. No.	Screening Question		Yes	No	Remarks
4.	Will the research work require interventions at field level?				
5.	Will the field based research work				
	i	located at or near an environmentally sensitive area?			
	ii	discharge any liquid waste in the environment?			
	iii	discharge large quantities of waste/used water?			
	iv	generate hazardous waste?			
	v	Impair downstream water quality?			
	vi	have any possible degradation in land and ecosystem?			
	vii	cause local air pollution from any plants/system operation?			
	viii	generate excessive noise and/or dust?			
6.	Will medical, biophysical or clinical research be conducted using human subjects?				
7.	Will the project have any indirect impact on environment and ecosystem?				
8.	Will the research work involve permission or clearance of any government department or agency?				
9.	Will future expansion or implementation of research finding cause any major environment problem?				

Annex 11 - Land Acquisition and Resettlement

Based on the findings of the Social Assessment Team' review of the EFA, Resettlement and or land acquisition problems are not anticipated in SSRP. However, there is always risk of exploitation and marginalization of Dalits, Women, marginalized and endangered Janjatis ¹ and other vulnerable groups by dominant groups. Nepalese society remains semi-feudal in character and the caste hierarchy, though officially illegal, remains very much in play in the rural areas. In order to ensure that the Dalits, Women, marginalized and endangered Janjatis and other vulnerable groups are not further marginalized in the course of development, it is recommended that the following approach be incorporated into the procedures for financing new classrooms or other school infrastructure of schools established during the SSRP period.

Rules of taking possession of land

Experience of implementing EFA reveals that Resettlement and or land acquisition problems are not anticipated in SSRP. The DoE will not do any involuntary land acquisition. All donations and purchases will be voluntary. DoE will not only ensure that any land donations are fully voluntary, but also that such donations do not involve physical displacement or any significant adverse impacts upon incomes of the donor household. All voluntary land transactions will meet the following criteria:

- (i) The land in question will be free of squatters, encroachers or other claims.
- (ii) Verification of the Voluntary nature of land donations in each case through formal public hearing.
- (iii) Land transfer will be completed through registration, with land title vested in the School; and
- (iv) Leasing of land or rights of use of land for schools will be duly documented.
- (v) Program Implementation Guidelines will include the principles of land acquisition described in this document.
- (vi) For any government building to be constructed under SSRP the land will be purchased from the open market. The DoE already has established practices in this regard and will follow the procedures while acquisition of land.

Procedure for school

Land Revenue office is responsible for land administration and registration of all types land in Nepal. The Survey Offices prepare land record through cadastral mapping. Major three types of land ownership are in practice i) registered private land, ii) registered public land and iii) un-registered (but cadastral mapped) government land (Ailani).

Normally the community approach to the land owner for contribution of land to the school. In this case either individual donate land free of cost or can ask for payment. To arrange the cash requirement there is practice that community people contribute cash to the school and school will make payment against the land take. These procedures are normally completed following willing sell or willing buy procedure of land administration.

¹ As per NEFI classification.

DEO will arrange for a public hearing when the schools established during the period of SSRP seeks grants for construction as per Rule XXXX. This public hearing shall verify the information provided in the land assessment data sheet. Minutes of the meeting will be made available at DEO office. If grievances/claims raised during public meeting are not settled during the public hearing the funds will not be granted to the school.

All land transactions will be subject to registration (as per market transactions) under the ownership of the School that is involved in the project and will be done only after ensuring that new school infrastructure or additional classroom construction will indeed be built in that particular piece of land.

A grievance redress mechanism will be in place to hear complaints regarding land acquisition.

and SMC are using the Ailani land for the establishment of the school. Such types of land are used by the schools with the communities or local body consent. In case of land requirement from the individual or group of people, school

Sample Formats

Land Acquisition assessment data sheet (to be used to record all land acquired for the schools)

- a) Location map

b) Land/ assets acquired

Plot No.	Area, local unit	Area, Sq.Ft equivalent	Description of assets in the land, if any	Date of acquisition (year, month, day)	Previous owner (choose one of the following: Individual/ Government/ Public)	Name of the owner [†]	Method of acquisition (choose one of the following: donation, sell-purchase/lease/rights of use)	Previous use (choose one of the following: barren land, farm land, squatter, describe in case of other uses)

c) Documentary proofs of land ownership or rights of use. (Notarized or witnessed statements)

[†] Provide name of the office in case of the government and name of the VDC in case of the public land.

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